GEOfinder

GSE228346, GSE173537, GSE211651, GSE154746, GSE202187, GSE127844, GSE66448, GSE89230, GSE112379, GSE144786, GSE209685

Experiment Type(s):

- [ ] Expression profiling by RT-PCR
- [ ] Expression profiling by array
- [ ] Expression profiling by high throughput sequencing
- [ ] Genome binding/occupancy profiling by genome tiling array

Species:

- [ ] Arabidopsis thaliana
- [ ] Bos taurus
- [ ] Caenorhabditis elegans
- [ ] Danio rerio
- [ ] Drosophila melanogaster
- [ ] Escherichia coli
- [ ] Gallus gallus
- [x] Homo sapiens

Number of Samples:

- [x] 1-10
- [x] 11-50
- [x] 51-100
- [x] 101-500
- [x] 501-1000
- [x] 1000+

Start Year:

End Year:

Submit

- [ ] Genome binding/occupancy profiling by high throughput sequencing
- [ ] Genome variation profiling by SNP array
- [ ] Genome variation profiling by genome tiling array
- [ ] Methylation profiling by array
- [ ] Methylation profiling by genome tiling array
- [ ] Methylation profiling by high throughput sequencing
- [ ] Non-coding RNA profiling by array
- [ ] Non-coding RNA profiling by high throughput sequencing
- [x] Other

- [ ] Mus musculus
- [ ] Oryza sativa
- [ ] Rattus norvegicus
- [ ] Saccharomyces cerevisiae
- [ ] Schizosaccharomyces pombe
- [ ] Sus scrofa
- [ ] synthetic construct

Top results (up to 100):

GSE ID	Title	Summary	Overa
GSE130758	Chromatin conformation remains stable upon extensive transcriptional	Heat shock (HS) causes dramatic and rapid changes in transcription of thousands of genes in metazoans. We have used	Investi chrom acute l human C. Two

GSE ID	Title	Summary	Overa
	changes driven by heat shock [K562]	insitu Hi-C in Drosophila S2 and human K562 cells to determine the effects of HS upon chromatin conformation. Our study shows that topologically associated domains (TADs), compartments and distal regulatory interactions remain unchanged upon HS.	were d (NHS) organi
GSE326884	The HSF2-HSP110 Axis Promotes Genome Stability by Supporting RNA Polymerase II Dependent Transcription and DNA Repair Gene Expression [PRO-Seq]	The precise regulation of RNA polymerase II (RNAPII) is essential for transcriptional fidelity and genome stability. Here, we identify a previously unrecognized genotoxic stressresponsive transcriptional axis composed of Heat Shock Factor2 (HSF2) and its client chaperone HSP110, which is activated by ionizing radiation (IR). Loss of HSF2 or HSP110 leads to increased DNA damage and heightened IR sensitivity. Mechanistically, the HSF2-HSP110 axis safeguards genome stability by sustaining RNAPII processivity and its C-terminal domain (CTD) phosphorylation, particularly at serine 7 (S7). Disruption of this axis leads to transcriptional dysregulation, altered pre-	Parent cells (i with ic werepe buffer HCl pH 250mM 1mM E (v/v) Ig (v/v) T and SU Subsec perfor Nascer at Har Martin

GSE ID	Title	Summary	Overa
		mRNA splicing, increased transcription-replication conflict, and downregulation of DNA damage response genes, resulting in DNA damage. In vivo, loss of HSF2 accelerates IR-induced lymphomagenesis by impairing transcriptional regulation, specifically inhibiting DNA repair gene expression. These findings define the HSF2-HSP110 axis as an essential transcriptional mechanism in the genotoxic stress response and reveal a therapeutic vulnerability that could be exploited to sensitize tumors to genotoxic therapies.	
GSE32060	Widespread regulation of translation by elongation pausing in heat shock	RNA-seq and ribosome footprinting libraries of mouse 3T3 and human 293T cellsrelated to Shalgi et al. 2013	36base and rib librarie Contro shock severe replica treated severe 3T3 ce inhibito et al. 2 transfe GFP, b of seve
GSE209687	Transcriptional responses of cancer	This SuperSeries is composed of the	Refer t

GSE ID	Title	Summary	Overa
	cells to heat shock-inducing stimuli involve amplification of robust HSF1 binding	SubSeries listed below.	
GSE200269	PRO-IP-seq: Pol II CTD code at nt-resolution in human K562 cells	Active sites of transcription were labelled with biotinylated nucleotide during a run-on reacation, where after an immunoprecipitation with antibodies that target Pol II CTD was conducted. This PRO-IP-seq protocol was conducted in human K562 erythroleukemia cells. The K562 cells were either untreated (NHS) or treated with 30 minutes of heat shock at 42C (HS).	human cells. T either treatec heat sh
GSE130892	MeRIP-seq for heat shock in B-cell lymphoma cells	We ran MeRIP-seq on RNA from heat shocked (43 C for 1 hr) and untreated cells to compare to previous reports as part of an evaluation of MeRIP-seq reproducibility.	3 repli and 3 OCI-Ly lympho
GSE234619	AFF4 globally affects the release of paused RNA polymerase II in HEL cells [PRO-seq]	P-TEFb, a heterodimer of kinase, CDK9, and Cyclin T1, is a critical regulator of promoter-proximal pause release of Pol II in metazoans. It is capable of forming three larger multiprotein complexes, including the super elongation complex (SEC), the BRD4/P-TEFb complex	One sa

GSE ID	Title	Summary	Overa
		and the 7SK snRNP. In the SEC or the BRD4/P-TEFb, P-TEFb is enzymatically active, while in the 7SK snRNP, its activity is inhibited. The SEC consists of AFF1 or 4, ENL or AF9, ELL1, 2 or 3 and EAF1 or 2 in addition to P-TEFb, the only subunit with catalytic activity, and the noncatalytic subunits have been found to be able to regulate pause release through P-TEFb. One recent study showed that in human DLD-1 cells, the SEC only regulates pause release of heat shock (HS) genes, whereas the BRD4/P-TEFb complex regulates pause release of the rest of the genes. In this study, we found that AFF4 knockdown in human HEL cells decreased not only cellular level but also global chromatin occupancy of CTD serine 2 phosphorylated Pol II, and notably, increased promoter-proximal pause of Pol II on several hundred HS and thousands of non-HS genes. Mechanistically, AFF4 facilitates pause release likely by facilitating the binding of P-TEFb to	

GSE ID	Title	Summary	Overa
		Pol II. These results suggest that the extent of impact of AFF4 on pause release is likely to be context-dependent or cell-type dependent.	
GSE101803	Oncogenes hijack the stress response machinery in T cell acute lymphoblastic leukemia [GRO-Seq]	Oncogenes NOTCH1 and MYC directly regulate HSF1 and other critical proteins of the stress-response pathway in T-ALL. This GRO-Seq experiment demonstrates that release from NOTCH1 inhibition results in upregulation of HSF1 and other key HSPs.	Nascer from 1 using T (Invitrc manuf follow (GRO) labelin contai using E which prepar sequer HiSeq read m was pe state (points NOTCH secret
GSE291973	HSF1 and YTHDC1 facilitate DNA damage repair upon heat stress by promoting the expressions of repair related genes	External stressors, especially heat stress, always lead to DNA damage and genome instability in cells. However, how cells rapidly response to these DNA damage remains largely unknown. In this study, we found that the major transcription factor in heat shock response, HSF1, activated several NHEJ	To inve m6A a NHEJ during we per not-he cells tr shocke treate U2OS siYTHD methyl immun

GSE ID	Title	Summary	Overa
		pathway related genes, especially NHEJ1, through the puncta it formed, thereby promoting the NHEJ pathway, reducing the DNA double-strand break upon heat stress. Furthermore, RNA m6A modification and its reader YTHDC1 were also found to play a role in this process, promoting the splicing of the NHEJ1 gene. Depletion of HSF1 or YTHDC1 led to increased nuclear H2AX intensities in the heat shocked cells, which was rescued by overexpression of NHEJ1. Besides, overexpression of HSF1 or YTHDC1 both caused increased NHEJ pathway level in the heat shocked cells, indicating the HSF1/YTHDC1 NHEJ1 DNA double-strand repair axis. Our findings investigate an approach through which cells repair their damaged DNAs in the heat shock response, offering a new insight of how cells maintain a bare-minimum genome stability when exposed to external stressors.	sequer U2OS
GSE70408	PAF1, a molecular regulator of promoter-proximal	The control of promoter-proximal pausing and the release of RNA	ChIP-s forms, subuni

GSE ID	Title	Summary	Overa
	pausing by RNA Polymerase II	polymerase II (RNA Pol II) is a widely used mechanism for regulating gene expression in metazoans, especially for genes that respond to environmental and developmental cues. Here, we identify Pol II associated Factor 1 (PAF1) as a major regulator of promoter-proximal pausing. Knockdown of PAF1 leads to increased release of paused Pol II into gene bodies at thousands of genes. Genes with the highest levels of paused Pol II exhibit the largest redistribution of Pol II from the promoter-proximal region into the gene body in the absence of PAF1. PAF1 depletion results in increased nascent transcription and increased levels of phosphorylation of Pol IIs c-terminal domain on serine 2 (Ser2P). These changes can be explained by the recruitment of the Ser2P kinase Super Elongation Complex (SEC) effecting increased release of paused Pol II into productive elongation, thus establishing a novel function for PAF1 as a	cell lin scramb total P targete scramb total P by Paf RNA-s and GF targete shRNA

GSE ID	Title	Summary	Overa
		major regulator of pausing in metazoans.	
GSE111904	Osmostress induced changes of chromatin architecture and transcription in mammalian cells [RNA-Seq, HiC]	Osmostress causes chromatin compaction and simultaneously triggers a massive transcriptional response in eukaryotic cells that is required for adaptation and cell survival. However, thus far it was unknown how changes of chromatin organization and ongoing DNA transcription are linked under hyper-osmotic conditions. In this study, we focused on elucidating the extend of chromatin reoganization upon hyper-osmotic shock, its reversibility and its link with active transcription in human T47D cells.	T47D c 100 m period RNA a collect RNA s betwe untrea compa h, 3 h NaCl. C transc occupa by Chl compa stresse used H organi NaCl t when i by pre nM of sequer recove chrom was as back t 1 h aft with ar additic CTCF I after 7 min co conditi ChIP-s evalua 1 h of stress

GSE ID	Title	Summary	Overa
			line wit compa
GSE105028	Architectural proteins and pluripotency factors cooperate to orchestrate the transcriptional response of hESCs to temperature stress	Cells respond to temperature stress via up- and down-regulation of hundreds of genes. This process is regulated by the heat shock factor HSF1, which controls the release of RNAPII from promoter-proximal pausing. Here we analyze the events taking place upstream of transcription elongation in hESCs. We find that temperature stress results in the activation or decommissioning of thousands of enhancers. This process involves alterations in the occupancy of transcription factors HSF1, AP-1, NANOG, KLF4, and OCT4, which is accompanied by changes in H3K27ac and H3K27me3, as well as nucleosome remodeling by BRG1. In addition, redistribution of RAD21 and CTCF results in the formation and disassembly of interactions mediated by these two proteins, either individually or in combination. These alterations either tether/untether enhancers to their cognate	Hi-C, H seq, el

GSE ID	Title	Summary	Overa
		promoters or remodel insulated neighborhoods with the ultimate goal of transforming the landscape of enhancer-promoter interactions. This loop remodeling process supports unidirectional loop extrusion as a mechanism for the establishment of CTCF/cohesin loops.	
GSE112608	Targeting Processive Transcription Elongation via SEC Disruption for MYC-Induced Cancer Therapy	The super elongation complex (SEC) is required for robust and productive transcription through release of RNA polymerase II (Pol II) with its P-TEFb module and promoting transcriptional processivity with its ELL2 subunit. Malfunction of SEC contributes to multiple human diseases including cancer. Here, we identify peptidomimetic lead compounds, KL-1 and its structural homolog KL-2, which disrupt the interaction between the SEC scaffolding protein AFF4 and P-TEFb, resulting in impaired release of Pol II from promoter-proximal pause sites and a reduced average rate of processive transcription elongation. SEC is required for induction of heat-shock	Determ affecti elonga II and S comple and wi or SEC

GSE ID	Title	Summary	Overa
		genes and treating cells with KL-1 and KL-2 attenuates the heat-shock response from Drosophila to human. SEC inhibition downregulates MYC and MYC-dependent transcriptional programs in mammalian cells and delays tumor progression in a mouse xenograft model of MYC-driven cancer, indicating that small-molecule disruptors of SEC could be used for targeted therapy of MYC-induced cancer.	
GSE130778	Chromatin conformation remains stable upon extensive transcriptional changes driven by heat shock	This SuperSeries is composed of the SubSeries listed below.	Refer t
GSE191242	HSP70 binds RNA and regulates transcription by RNA Polymerase III [FLASH-seq]	We examined the RNA binding ability of the 70-kDa heat shock protein (HSP70) family member HSPA1A and HSP8 and identified the RNA species bound to HSP70 or HSP8 in vivo. We used a UV-crosslinkiung and immunoprecipitation based approach named FLASH-seq (Akta? et al., 2017) using 3xFLAG-Histidine-Biotin tagged	FLASH tagged HSPA8 replica

GSE ID	Title	Summary	Overa
		ectopically expressed protein in HEK293T cells.	
GSE159633	RTF1 and SPT6 stimulate transcription elongation by two distinct mechanisms	We have previously shown that the RNA polymerase II (Pol II)-DSIF complex associates with the PAF1 complex (PAF), RTF1 and SPT6 to form an activated elongation complex in vitro. Here we investigate the mechanisms that these factors use to stimulate Pol II elongation in vivo. We combine rapid factor depletion from human cells with genome-wide analyses of changes in RNA synthesis and occupancy with engaged Pol II. Whereas depletion of PAF subunits has little effect on transcription in vivo, depletion of RTF1 or SPT6 strongly compromises RNA synthesis, albeit in fundamentally different ways. RTF1 depletion decreases Pol II velocity, whereas SPT6 depletion impairs Pol II progression through nucleosomes. These results show that distinct transcription elongation factors stimulate Pol II velocity and Pol II progression through chromatin in vivo. Our results also provide evidence for two distinct	Sample were c duplica with dT the co were t duratic (DMSC dilutior

GSE ID	Title	Summary	Overa
		barriers to elongation at the beginning of genes, the promoter-proximal pause site and the +1 nucleosome.	
GSE192848	Coordinated regulation of RNA polymerase II pausing and elongation progression by PAF1	Pleiotropic transcription regulator RNA polymerase II (Pol II)-associated factor 1 (PAF1) governs multiple transcriptional steps and the deposition of several epigenetic marks. However, it remains unclear how ultimate transcriptional outcome is determined by PAF1 and whether it relates to PAF1-controlled epigenetic marks. We utilize rapid degradation systems and reveal direct PAF1 functions in governing pausing partially by recruiting Integrator-PP2A (INTAC), in addition to ensuring elongation. Following acute PAF1 degradation, destabilized polymerase undergoes early termination or effective release, which presumably relies on skewed balance between INTAC and P-TEFb resulting in hyperphosphorylated substrates including SPT5. Impaired Pol II progression during elongation, along with altered pause release	We pe seq, or the rol transc sample were c duplica with d the co were t duratic (DMSC dilutior

GSE ID	Title	Summary	Overa
		frequency, determines the final transcriptional outputs. Moreover, PAF1 degradation causes a cumulative decline in histone modifications. These epigenetic alterations in chromatin likely further influence the production of transcripts from PAF1 target genes.	
GSE295212	Compromised Transcriptional Elongation Stimulates Initiation in Human Cells [4sU-DRB-seq]	P-TEFb is a well-established positive regulator of pause release of Pol II. The human PAF1 complex (PAF1C), consisting of PAF1, CTR9, CDC73, CTR9 and SKI8, plays critical roles in transcription and RNA processing, yet whether and how it regulates pause release is under debate. To address these issues, we further analyzed effects of PAF1 loss on pause release while using P-TEFb inhibition as a positive control. We found that besides increasing pause of Pol II, P-TEFb inhibition decreases and increases chromatin occupancy of PAF1C and Integrator, respectively. In contrast, acute PAF1 degradation increases the occupancy of Integrator and ARMC5, suggesting a positive role of the PAF1C	10 mill each 4 To det reduct output produc dTAG 100 M clear e gene b DMSO the ad acute Subsec washe PBS to effect were t thiouri for 0, respec

GSE ID	Title	Summary	Overa
		in pause release by preventing promoter-proximal termination and Pol II degradation. Moreover, combined P-TEFb inhibition and PAF1 degradation increase pause of Pol II more than either treatment alone does, suggesting a cooperativity between them in pause release. Importantly, either P-TEFb inhibition or PAF1 degradation increases occupancy of initiation factors, suggesting that elongation perturbation may stimulate initiation, contrary to conclusions of three previous studies. Altogether, our results firmly established the PAF1C as a critical positive regulator of pause release and advanced current understanding of the connection between pause and initiation.	
GSE100742	Oxidative stress elicits a rapid and global RNA polymerase II transcriptional response in human cells	In GSE93931, we used ChIP-Seq to show that hydrogen peroxide causes rapid accumulation of Pol II at promoters and enhancers. Over time, a fraction of accumulated Pol II loses NELF and creeps downstream. In GSE100740, we also compared nuclear walk-on	ChIP-S Seq us cells | Series

GSE ID	Title	Summary	Overa
		results with PRO-Seq, both of which were performed using nuclei from DMSO- and flavopiridol-treated cells. | This SuperSeries is composed of the SubSeries listed below.	
GSE96869	Nascent RNA sequencing reveals a dynamic global transcriptional response at genes and enhancers to the natural medicinalcompound celastrol	Most studies of responses to transcriptional stimuli measure changes incellular mRNA concentrations. By sequencingnascent RNA instead, it is possible to detect changes in transcription inminutes rather than hours, and thereby distinguish primary from secondary responses to regulatory signals. Here, we describe the use of PRO-seq to characterize the immediate transcriptional response in human cells to celastrol, a compound derived from traditional Chinese medicine that has potent anti-inflammatory, tumor-inhibitory and obesity-controlling effects. Our analysis of PRO-seq data for K562 cells reveals dramatic transcriptional effects soon after celastrol treatment at a broad collection of both coding and noncoding transcription units. Thistranscriptional	Sample 2 repli

GSE ID	Title	Summary	Overa
		response occurred in two major waves,one within 10 minutes, and a second 40-60 minutes after treatment. Transcriptional activity was generally repressed by celastrol,but one distinctgroup of genes, enriched for rolesinthe heat shock response, displayed strong activation. Using a regression approach, we identified key transcription factors that appear to drive these transcriptional responses, including members of the E2F and RFX families. We also found sequence-basedevidence that particular TFs drive the activation of enhancers. We observed increased polymerase pausing at both genes and enhancers, suggesting that pause release may be widely inhibited during the celastrol response. Our study demonstrates that a careful analysis of PRO-seq time coursedata can disentangle key aspects of acomplex transcriptional response, and it provides new insights into the activity of a powerful pharmacological agent.	

GSE ID	Title	Summary	Overa
GSE164200	m6A-RIP-seq of heat shock-exposed HeLa cells upon HSATIII knockdown	To investigate the effect of HSATIII lncRNA on m6A modification, we performed m6A-RIP(RNA immuno precipitation) RNA-seq from heat shock-exposed HeLa cells upon HSATIII knockdown.	We pe non-st expose withou knock
GSE269247	Kinetic measurement of gene-specific RNAPII transcription elongation rates	Transcription is regulated at multiple levels, including initiation, elongation, and termination. Whereas much research has focused on initiation of transcription, regulation of elongation plays an important role not only for transcription dynamics, but also for co-transcriptional RNA processing as the RNAPII elongation rate can influence maturation of the nascent RNA transcript. Despite advances in high-throughput approaches for global identification of RNAPII speed, RNAPII elongation rate studies have been largely limited to a small number of genes. Here, we present DRB/TTchem-seq2, a modified version of DRB/TTchem-seq method, to measure gene-specific elongation rates of more than 3,000 genes. By	Nascer RNA P release proxim captur sequer time p 20 mir DLD-1 rates v the dis RNAPI for bot cells.

GSE ID	Title	Summary	Overa
		approximating the distance traveled by RNAPII in certain time intervals using the wave fronts post RNAPII release. Our direct RNAPII elongation rate quantification reveals that elongation rates vary not only among genes but also within genes. Additionally, specific histone modifications and elongation factors are tightly correlated with the elongation rates. Together, we present a robust and powerful method for RNAPII transcription elongation rate measurement.	
GSE100740	Oxidative stress elicits a rapid and global RNA polymerase II transcriptional response in human cells [PRO-Seq]	We compared nuclear walk-on results with PRO-Seq, both of which were performed using nuclei from DMSO- and flavopiridol-treated cells. In GSE93931, we also used ChIP-Seq to show that hydrogen peroxide causes rapid accumulation of Pol II at promoters and enhancers. Over time, a fraction of accumulated Pol II loses NELF and creeps downstream.	PRO-S HeLa c
GSE242378	FACT orchestrates the interplay between chromatin structure	Transcription by RNA polymerase II (RNA Pol II) depends on transcription	Sampl were c duplic

GSE ID	Title	Summary	Overa
	and transcription [5Cap-Seq]	factors and chromatin factors. Here we use rapid factor depletion and multiomics analysis to investigate how a histone chaperone, FAcilitates Chromatin Transcription (FACT), influence nascent transcription by RNA PolII in human cells. Depletion of a FACT subunit, SSRP1, led to rapid changes in chromatin structure and concomitantly strongly compromised RNA synthesis. FACT depletion led to a multilayered transcriptional defect, including loss of promoter proximal pausing, deregulated release into elongation and drop-off of RNA Pol II in promoter-distant gene regions. We combined these analyses with biochemical dissection of transcription of a chromatinized template to show that FACT supports both elongation and pausing of RNA Pol II. Our study also provides new evidence how the position of promoter proximal pausing is defined by the +1 nucleosome in human cells.	with d the co were t duratic (DMSC dilutior

GSE ID	Title	Summary	Overa
GSE148844	Elongation factor ELOF1 drives transcription-coupled repairand prevents genome instability [DRB-TTchem-seq]	Correct transcription is crucial for life. However, DNA damage severely impedes elongating RNA Polymerase II (Pol II), causing transcription inhibition and transcription-replication conflicts. Cells have evolved intricate mechanisms to counteract the severe consequence of these transcription-blocking lesions, however the exact mechanism and factors involved remain largely unknown. Here, using a genome-wide CRISPR/cas9 screen, we identified elongation factor ELOF1 as an important new factor in the damage-induced transcription stress response. ELOF1 has an evolutionary-conserved role in TC-NER where it promotes the recruitment of the TC-NER factors UVSSA and TFIIH and facilitates Pol II ubiquitylation to efficiently repair TBLs. Importantly, ELOF1 has an additional role in protecting cells from transcription-mediated replication hindrance, thereby preserving genome stability. Thus, ELOF1	Determ rates i KO, CS ELOF1 TTcher transc seq); 3 with 1(1--D- ribofur followe inhibiti min (in thiouri

GSE ID	Title	Summary	Overa
		protects the transcription machinery from DNA damage by two distinct mechanisms.	
GSE140930	A CSB-PAF1C axis restores processive transcription elongation after DNA damage repair	The coordinated transcription of genes involves the regulated release of RNA polymerase II (RNAPII) from promoter-proximal sites into active elongation. DNA lesions in transcribed strands block elongation and induce a strong transcriptional arrest. The transcription-coupled repair (TCR) pathway efficiently removes transcription-blocking DNA lesions, but this is not sufficient to resume transcription. Through proteomics screens, we find that the TCR-specific CSB protein loads the evolutionary conserved PAF1 complex (PAF1C) onto RNAPII in promoter-proximal regions in response to DNA damage. PAF1C is dispensable for TCR-mediated repair, but is essential for recovery of RNA synthesis after UV irradiation, suggesting an uncoupling between DNA repair and transcription recovery. Moreover, we find that PAF1C promotes	BrU-se human untrea 26h af varyin sample perfor PAF1 i knock U2OS deplet induci seq wa U2OS auxin-i PAF1.

GSE ID	Title	Summary	Overa
		RNAPII pause release in promoter-proximal regions and subsequently acts as a processivity factor that stimulates transcription elongation throughout genes. Our findings expose the molecular basis for a non-canonical PAF1C-dependent pathway that restores transcription throughout the human genome after genotoxic stress.	
GSE38140	GRO-seq from HCT116 cells	The majority of transcription studies examine steady-state RNA . However steady-state RNA is not a true reflection of the transcriptome, because the RNA levels are affected by both transcription rate and degradation rate. In this experiment we measured the amount of transcription occurring in HCT116 colon cancer cells, regardless of degradation, using GRO-seq (global nuclear run-on sequencing). This information demonstrates that many genes have a pile-up of transcriptionally-engaged polymerase near their 5'-end.	Nuclei HCT11 with DI additic experi separa on was Core, I Lis, J.T sequer pausin initiatic Scienc nascer and se

GSE ID	Title	Summary	Overa
GSE181161	Quantifying RNA Synthesis at Rate-Limiting Steps of Transcription Using Nascent RNA-sequencing Data	Precision Run-On sequencing (PRO-seq) data from human K562 cells, mapped to hg38.	The ra genera (Viher Cell; G PRO-s unstre remap genom
GSE166202	STL-seq reveals distinct pause release and termination kinetics for promoter-proximal paused RNA polymerase II transcripts	Despite the critical regulatory function of promoter-proximal pausing, the influence of pausing kinetics on transcriptional control remains an active area of investigation. Here, we present Start-TimeLapse-seq (STL-seq), a method that captures the genome-wide kinetics of short, capped RNA turnover and reveals principles of regulation at the pause site. By measuring the rates of release into elongation and premature termination through inhibition of pause release, we determine that pause-release rates are highly variable and most promoter-proximal paused RNA Polymerase II molecules prematurely terminate (~80%). The preferred regulatory mechanism upon a hormonal stimulus (20-hydroxyecdysone) is to	RNA is S2 cell hormo inhibit triptoli subjec nucleo substit enrichi RNA s and en RNA f nucleo substit isolate cells tr s4U w oxidati aroma chemis short, Seque prepar

GSE ID	Title	Summary	Overa
		influence pause-release rather than termination rates. Transcriptional shutdown occurs concurrently with induction of promoter-proximal termination under hyperosmotic stress but paused transcripts from TATA box-containing promoters remain stable, demonstrating an important role for cis-acting DNA elements in pausing. STL-seq dissects the kinetics of pause release and termination, providing an opportunity to identify mechanisms of transcriptional regulation.	
GSE61332	Native Elongating Transcript Sequencing Reveals Human Transcriptional Activity at Nucleotide Resolution	Major features of transcription by human RNA polymerase II (Pol II) remain poorly defined due to a lack of quantitative approaches for visualizing Pol II progress at nucleotide resolution.We developed a simple and powerful approach for performing native elongating transcript sequencing (NET-seq) in human cells that globally maps strand-specific Pol II density at nucleotide resolution. NET-seq exposes a mode of antisense transcription that originates	10 san NET-se consis replica HeLaS conditi contro conditi NET-se cells c biologi (HEK2 HEK29 data fc consis fractio (HeLaS HeLaS and HeLaS

GSE ID	Title	Summary	Overa
		downstream and converges on transcription from the canonical promoter. Convergent transcription is associated with a distinctive chromatin configuration and is characteristic of lower-expressed genes. Integration of NET-seq with genomic footprinting data reveals stereotypic Pol II pausing coincident with transcription factor occupancy. Finally, exons retained in mature transcripts display Pol II pausing signatures that differ markedly from skipped exons, indicating an intrinsic capacity for Pol II to recognize exons with different processing fates.	RNA-s cells c cytopl conditi (HEK2
GSE233930	NELF Focuses Sites of Initiation and Maintains Promoter Architecture	Many factors control the elongation phase of transcription by RNA polymerase II (Pol II), a process that plays an essential role in regulating gene expression. We utilized cells expressing degradation tagged subunits of NELFB, PAF1 and RTF1 to probe the effects of depletion of the factors on nascent transcripts using PRO-Seq and on chromatin architecture using DFF-	To inve NELFB create which for ea tagge This al rapidly these addit NELFB and 4 absen RTF1 PAF1 perfor

GSE ID	Title	Summary	Overa
		ChIP. Although NELF is involved in promoter proximal pausing, depletion of NELFB had only a minimal effect on the level of paused transcripts and almost no effect on control of productive elongation. Instead, NELF depletion increased the utilization of downstream transcription start sites and caused a dramatic, genome-wide loss of H3K4me3 marked nucleosomes. Depletion of PAF1 and RTF1 both had major effects on productive transcript elongation in gene bodies and also caused initiation site changes like those seen with NELFB depletion. Our study confirmed that the first nucleosome encountered during initiation and early elongation is highly positioned with respect to the major TSS. In contrast, the positions of H3K4me3 marked nucleosomes in promoter regions are heterogeneous and are influenced by transcription. We propose a model defining NELF function and a general role of the H3K4me3	quanti genom transc Pol-II a Seq or dTAGV Pol-II e cells fc H3K4r deplet Finally, genom cells w Sf21 n lines w contro Seq, re biologi sample (two di

GSE ID	Title	Summary	Overa
		modification in blocking transcription initiation.	
GSE188410	The PAF1 Complex Regulates Transcriptional Termination and Reinitiation [4sUDRB-seq]	Chromatin places fundamental physical constraints on transcription (Gamarra and Narlikar, 2021). The PAF1 complex (PAF1C), a hexamer of PAF1, LEO1, CTR9, SKI8, CDC73 and RTF1, plays a critical role in transcription with incompletely understood mechanisms. During transcriptional elongation, PAF1C is one of the positive elongation factors in complex with RNA polymerase II (Pol II) to facilitate elongation through chromatin (Vos et al., 2018), but it is unclear yet if negative elongation factors are needed concurrently to restrain elongation. Here we show that besides decreasing elongation rate, LEO1 knockout in human K562 cells increases transcriptional readthrough and cellular level of C-terminal domain (CTD) phosphorylated Pol II while increases and decreases transcriptional output of several hundred genes, respectively. Mechanistic analyses taking proteomic,	4sUDF KO and K562 c

GSE ID	Title	Summary	Overa
		functional genomic and biochemical approaches discovered that PAF1C regulates transcriptional termination in part through recruiting PNUTS-PP1 complex and facilitates Pol II transcriptional re-initiation through recruiting TOX4-PP1 complex. Moreover, Paf1 conditional knockout in mice severely blocks T cell development, increases cellular level of CTD phosphorylated Pol II, mainly decreases Pol II occupancy and transcriptional output in double positive T cells, and importantly, the regulation of re-initiation by PAF1C through TOX4-PP1 complex is conserved between mouse and human. Our results also suggest that PNUTS-PP1 and TOX4-PP1 bind PAF1C to restrain Pol II elongation through chromatin. These findings not only establish PAF1C as a critical regulator of transcriptional termination and re-initiation besides elongation but also advanced current understanding of elongation.	

GSE ID	Title	Summary	Overa
GSE141373	DRB-PRO-Seq for measuring Pol II transcriptional elongation rate alterations upon inhibition of analog sensitive CDK12 and/or CDK13	We performed PRO-Seq to determine Pol II elongation rates by measruing the distance of Pol II travels during 25 minutes after Pol II being relased from DRB induced pausing. We allowed Pol II to release in the presence of 1-NM-PP1 or DMSO after DRB being wahsed out in order to assess the effects of CDK12 and/or CDK13 inhibtion on transcriptional elongation rates.	Cells w DRB fo was wa of DRE elonga cells w minute DMSO after 2 we me Pol II tı minute to calc
GSE166658	K562 PRO-seq	Nascent transcription measured by PRO-seq in untreated and DMSO treated conditions.	To gen nascer cells al two PF were p
GSE139237	A Unified View of the Sequence and Functional Organization of the Human RNA Polymerase II Promoter	To expand our understanding of the interplay of human RNA polymerase II (Pol II) promoter elements, promoter-proximal pausing, local chromatin structure, and co-transcriptional RNA processing, the 5 and 3 ends of nascent, capped transcripts and the locations of nearby nucleosomes were accurately identified. High-quality visualization tools revealed sequence elements defining the core promoter, sites of pausing,	NasCa protoc end to followe cappe m7G o cappe DFF-Se nucleo the hu estima Nuclea with DI Howev occupa from tl nuclea Factor investi

GSE ID	Title	Summary	Overa
		and nucleosome positioning across over 177,000 transcription start regions (TSRs). We found that cap methylation begins as transcripts attain a length of about 30 nt. A nuclear run-off assay was developed that utilizes the unique properties of the DNA fragmentation factor (DFF) to demonstrate the relative location of nucleosomes and paused Pol II. Our data support the conclusion that human Pol II promoters are functional TFIID binding sites with built-in downstream information directing ubiquitous promoter proximal pausing and the downstream nucleosome location.	our ow nucleo DFF ar sequer treatec inhibit nuclei treatec have s undige cells tr flavopi 30 mir
GSE311646	EnChAMP-seq genomic localization and footprinting of RNA Pol II and NELF-A	RNA Polymerase II (Pol II) transcription is highly regulated at two early steps in the transcription cycle: 1) Pre-Initiation Complex (PIC) assembly with its coupled initiation, and 2) promoter-proximal pausing with its controlled release. Here, we developed an optimized biochemical purification method that captures endogenously tagged	EnChA endog (contr NELFA cell lin no dru 1h, or 20min

GSE ID	Title	Summary	Overa
		chromatin-bound Pol II complexes under native conditions at these rate-limiting steps. We then identified a large set of Pol II interactors by mass spectrometry and determined the footprints of these assemblies on promoters with high resolution. Many well-known and new or understudied factors were identified as associated with the PIC and promoter proximal paused complexes, indicating that despite decades of efforts, these rate-limiting steps of the transcription cycle are far from being completely understood. The new and understudied factors implicate novel mechanisms of regulation that will need to be characterized to fully understand Pol II regulation.	

GSE ID	Title	Summary	Overa
GSE244353	Tyrosine 1 phosphorylated RNA polymerase II transcribes PROMPTs to promote proximal promoter pausing and induce global transcriptional repression in response to DNA damage	DNA faces continuous threats from various endogenous and exogenous stresses. When damaged, DNA requires repair to enable proper cell functioning. Cells have evolved multiple repair pathways to maintain genome stability. Transcription mediated by RNA polymerase II (RNA Pol II), which is conventionally linked with gene expression, plays an integral role in DNA repair. Transcription occurring at sites of DNA damage facilitates repair by generating functional transcripts, recruiting DNA repair factors, or modulating chromatin architecture. Conversely, the shutdown of transcription in nearby genes is crucial to ensure effective DNA repair. UV-induced DNA lesions have been shown to lead to a genome-wide suppression of transcription through RNA Pol II ubiquitylation and subsequent degradation. However, the global transcriptional response to double-strand breaks remains unclear.In this study, we employed mNET-Seq analysis to	mNet-S using T phospl II, Serir phospl II,Serin phospl II antib either irradia antibo RNA P using r (Nojim nascer bound

GSE ID	Title	Summary	Overa
		examine nascent transcripts bound to the CTD Y1P, S2P and S5P of RNA Pol II. Our data reveal a global transcriptional shutdown, as evidenced by a greater number of downregulated protein coding genes in response to irradiation. Interestingly, we observed a significant increase in Y1P antisense transcripts near promoters, known as PROMPTs, which are associated with the downregulated protein coding genes upon irradiation. Furthermore, an increased promoter pausing index correlated with elevated expression of PROMPTs in response to double strand breaks induction. Motif enrichment analysis uncovered binding sites for the Polycomb repressive complex 2 (PRC2) in upregulated PROMPTs. Using proximity ligation assays, we demonstrated an increased proximity between Y1P RNA Pol II and PRC2 upon irradiation. Finally, we also report that inhibition of Y1P or depletion of PRC2 results in the loss of	

GSE ID	Title	Summary	Overa
		transcriptional repression of protein coding genes, associated with PROMPTs.	
GSE69141	7SK-BAF axis controls pervasive transcription at enhancers [ChIRP-Seq]	Eukaryotic genomes are extensively transcribed, but unfettered transcription alters gene expression and leads to genome damage by several means. Divergent transcription occurs at active enhancers and promoters, distinct classes of cis-regulatory elements critical for precise control of gene expression. A key step in RNA Polymerase II (Pol II) transcription is promoter-proximal pausing, which occurs bidirectionally ~25-60 nucleotides downstream of transcription start sites (TSS). Promoter-proximal pause release is gated by the positive transcription elongation factor b (P-TEFb)-7SK snRNA pathway; release from 7SK allows P-TEFb phosphorylation of Pol II and subsequent elongation. The 7SK small nuclear ribonucleoprotein (snRNP) is thought to reside in the nucleoplasm, but it has been suggested that 7SK could operate physically on chromatin. Notably, while enhancer	Each e perfor duplic contro

GSE ID	Title	Summary	Overa
		transcription is one of the earliest steps of gene activation12 and some enhancer RNAs (eRNAs) participate in gene regulation, far less is known about the control of eRNA transcription. Here we show that 7SK inhibits enhancer transcription by modulating nucleosome position. 7SK occupies enhancers and super enhancers genome-wide, and 7SK is required to limit eRNA initiation and synthesis in a manner distinct from promoter pausing. Clustered elements at super enhancers uniquely require 7SK to prevent convergent transcription of colliding polymerases. 7SK inhibits enhancer transcription by modulating chromatin structure, physically interacts with the BAF chromatin remodeling complex, and is required to recruit BAF to enhancers. Thus, 7SK employs distinct mechanisms to counteract diverse consequences of pervasive transcription that distinguish super	

GSE ID	Title	Summary	Overa
		enhancers, enhancers, and promoters	
GSE218726	Modeling the nascent RNA transcriptome with chrTT-seq to extract chromatin dissociation dynamics on newly transcribed RNA.	To capture chromatin dissociation dynamics of nascent RNA transcripts we labeled MCF-7 cells with 4-thio-uridine (4-SU) for 8 minutes, then chased with an excess of uridine for 5, 10, 15 and 20 min. At each pulse-chase time point we performed TT-seq from the chromatin-associated and nucleoplasmic fraction. | The ratios of spike-in normalized chromatin-associated to chromatin-released nascent RNA read coverage over genomic features (transcripts' last exons) were fitted on an exponential decay function to extract an estimated chromatin association half-live time for each transcript.	Briefly, was pe RNA w phenol associ release chemic NaOH based transc three 4 unlabe added fragme Illumin constr TrueSe Library
GSE307864	Maintenance of gene expression homeostasis by systems-level feedback loops in response to changes in RNA polymerase II dosage [RNAPII_levels_DRB-TTseq]	Transcription is regulated by sequence-specific transcription factors and enzymes allowing access to genes in chromatin. However, recent data indicate that the abundance of RNA polymerase II (RNAPII) may itself under certain circumstances represent	To inve RPB1 elonga RPB1 degror concer After was sy promo throug Afterw

GSE ID	Title	Summary	Overa
		an additional, crucial determinant of transcription regulation. Here, we titrated the cellular dosage of human RPB1, the largest RNAPII subunit, using the dTAG system to more generally assess the importance of RNAPII levels. Unexpectedly, cells are extremely sensitive to RPB1 dosage, with a mere 30% reduction sufficient to perturb cell proliferation, cell cycle progression, and global transcription. Using a combination of sequencing and proteomic approaches, we uncover the existence of multiple feedback loops between transcriptional initiation, promotor-proximal pause release, transcript elongation, splicing, and mRNA half-life, which facilitate the maintenance of gene expression homeostasis. Together, our data suggests that gene expression is highly sensitive to RPB1 dosage and that the cellular levels of RNAPII must be tightly regulated.	were r times nascer labelec the uri (4sU). chem- seq) w fragme sequer high-re nascer the tra was m points speed.
GSE188417	The PAF1 Complex Regulates Transcriptional	Chromatin places fundamental physical constraints on	Refer

GSE ID	Title	Summary	Overa
	Termination and Reinitiation	transcription (Gamarra and Narlikar, 2021). The PAF1 complex (PAF1C), a hexamer of PAF1, LEO1, CTR9, SKI8, CDC73 and RTF1, plays a critical role in transcription with incompletely understood mechanisms. During transcriptional elongation, PAF1C is one of the positive elongation factors in complex with RNA polymerase II (Pol II) to facilitate elongation through chromatin (Vos et al., 2018), but it is unclear yet if negative elongation factors are needed concurrently to restrain elongation. Here we show that besides decreasing elongation rate, LEO1 knockout in human K562 cells increases transcriptional readthrough and cellular level of C-terminal domain (CTD) phosphorylated Pol II while increases and decreases transcriptional output of several hundred genes, respectively. Mechanistic analyses taking proteomic, functional genomic and biochemical approaches discovered that PAF1C regulates transcriptional termination in part	

GSE ID	Title	Summary	Overa
		through recruiting PNUTS-PP1 complex and facilitates Pol II transcriptional re-initiation through recruiting TOX4-PP1 complex. Moreover, Paf1 conditional knockout in mice severely blocks T cell development, increases cellular level of CTD phosphorylated Pol II, mainly decreases Pol II occupancy and transcriptional output in double positive T cells, and importantly, the regulation of re-initiation by PAF1C through TOX4-PP1 complex is conserved between mouse and human. Our results also suggest that PNUTS-PP1 and TOX4-PP1 bind PAF1C to restrain Pol II elongation through chromatin. These findings not only establish PAF1C as a critical regulator of transcriptional termination and re-initiation besides elongation but also advanced current understanding of elongation.	

GSE ID	Title	Summary	Overa
GSE277764	Nanopore long-read RNA sequencing in untreated (UN), sodium arsenite (SA) and heat shock (HS) stressed HeLa cells	Recent completion of the telomere-to-telomere (T2T) genome assembly has enabled a comprehensive characterization of pericentromeric Sat, SatII, Sat and centromeric -satellite repeats. Sat DNA constitutes ~1.56% of the genome with a reported localization across 16 chromosomes. The transcription activity of Sat DNA across genome and the sequence of Sat transcripts remained largely unclear. We performed nanopore long-read RNA sequencing (RNA-seq) in untreated (UN), sodium arsenite (SA: 0.1mM, 5 h) and heat shock (HS: 42C, 2 h; 37C, 1 h) stressed HeLa cells to characterize Sat transcripts . Since a portion of Sat transcripts is non-polyadenylated, We performed polyadenylated (poly(A)+) and rRNA-depleted (ribo-) nanopore cDNA long-read RNA-seq.	We pe (poly(A (ribo-) RNA-s sodium shock cells. T indepe each c
GSE191245	HSP70 binds to specific non-coding RNA and regulates human RNA polymerase III	This SuperSeries is composed of the SubSeries listed below.	Refer t

GSE ID	Title	Summary	Overa
GSE188416	The PAF1 Complex Regulates Transcriptional Termination and Reinitiation [TT-seq]	Chromatin places fundamental physical constraints on transcription (Gamarra and Narlikar, 2021). The PAF1 complex (PAF1C), a hexamer of PAF1, LEO1, CTR9, SKI8, CDC73 and RTF1, plays a critical role in transcription with incompletely understood mechanisms. During transcriptional elongation, PAF1C is one of the positive elongation factors in complex with RNA polymerase II (Pol II) to facilitate elongation through chromatin (Vos et al., 2018), but it is unclear yet if negative elongation factors are needed concurrently to restrain elongation. Here we show that besides decreasing elongation rate, LEO1 knockout in human K562 cells increases transcriptional readthrough and cellular level of C-terminal domain (CTD) phosphorylated Pol II while increases and decreases transcriptional output of several hundred genes, respectively.	TT-seq transcr contro KO hur WT an thymo

GSE ID	Title	Summary	Overa
		Mechanistic analyses taking proteomic, functional genomic and biochemical approaches discovered that PAF1C regulates transcriptional termination in part through recruiting PNUTS-PP1 complex and facilitates Pol II transcriptional re-initiation through recruiting TOX4-PP1 complex. Moreover, Paf1 conditional knockout in mice severely blocks T cell development, increases cellular level of CTD phosphorylated Pol II, mainly decreases Pol II occupancy and transcriptional output in double positive T cells, and importantly, the regulation of re-initiation by PAF1C through TOX4-PP1 complex is conserved between mouse and human. Our results also suggest that PNUTS-PP1 and TOX4-PP1 bind PAF1C to restrain Pol II elongation through chromatin. These findings not only establish PAF1C as a critical regulator of transcriptional termination and re-initiation besides elongation but also advanced current	

GSE ID	Title	Summary	Overa
		understanding of elongation.	
GSE132764	TFIID enables RNA polymerase II promoter-proximal pausing	RNA polymerase II (pol II) transcribes all protein-coding and many non-coding RNAs in the human genome. Pol II transcription initiation is governed by the Pre-Initiation Complex (PIC), which contains TFIIA, TFIIB, TFIID, TFIIE, TFIIF, TFIIH, pol II, and Mediator. After initiation, pol II enzymes typically pause after transcribing less than 100 bases, and paused polymerases represent a common regulatory intermediate. Accordingly, paused pol II has been implicated in enhancer function, development and homeostasis, and diseases ranging from cancer to viral pathogenesis. Precisely how pol II promoter-proximal pausing is enforced and regulated remains unclear; however, protein complexes such as NELF and DSIF increase pausing whereas the activity of CDK9 (P-TEFb complex) correlates with pause release. To address specific mechanistic questions about pol II pausing and its regulation,	Examir by per sequer HCT11 TAF1 i contro

GSE ID	Title	Summary	Overa
		we reconstituted human pol II promoter-proximal pausing in vitro, entirely with purified factors (no extracts). As expected, NELF and DSIF increased pol II pausing in vitro, whereas P-TEFb promoted pause release. Unexpectedly, the PIC alone was sufficient to reconstitute pol II pausing, suggesting that pausing is an inherent property of the PIC. In agreement, pol II pausing was lost upon replacement of the TFIID complex with TATA-binding protein (TBP); moreover, pausing was dependent upon TFIID subunits TAF1 and TAF2. TAF1/2 bind genomic DNA downstream of the pol II initiation site, invoking a complex interaction model for pausing. Consistent with this model, PRO-Seq experiments revealed increased transcription upon acute depletion (t=60 min) of TAF1 and TAF2 in human cells, and pol II pausing was disrupted at thousands of genes. Similar results were obtained in TAF1-depleted Drosophila S2 cells. Collectively, these data establish the general	

GSE ID	Title	Summary	Overa
		transcription factor TFIID as a genome-wide regulator of pol II promoter-proximal pausing.	
GSE290558	Mapping breakome of multiple cancer cell-lines	Cancer is characterized by uncontrolled proliferation accompanied by the hypertranscription of oncogenes, leading to transcription stress, a key source of DNA double-strand breaks (DSBs) that jeopardize genomic stability. Yet, transcription stress is still underexplored. In this study, we utilized maps of DSBs identified through in-suspension break labeling in situ and sequencing (sBLISS), along with transcription stress markers, revealing that transcription stress regions coincide with the super-enhancer regulatory landscape. Notably, H2AX mapping indicates its enrichment at transcription stress sites while not all DSB-enriched genes show equal H2AX marking, but those with DSBs tied to transcription stress are distinctly marked. Intriguingly, genes with high-DSBs marked by H2AX exhibited significantly higher DSB	MCF7 (HTB-1 RPMI s (vol/vc FBS (G penicill 293 (C (HTB-1 (HTB-2 2) were supple (vol/vc penicill cells (H EMEM FBS, g penicill SY5Y (in DME with 1(penicill cells w mamm media supple whereа on DM with 5(ng/ml Hydroc Choler and Pe were s cell cy(serum-

GSE ID	Title	Summary	Overa
		turnover and repair than those with H2AX-low genes, manifesting vulnerability to mutagenesis. These findings underscore super-enhancer activity as a determinant of the transcription stress landscape in cancer, posing a threat to the genomic stability of oncogenes in cancer.	followe media 5 hour 37C ur atmosp Cells w auther and te: and ce passag were tr (Sigma (Sigma CDK9i 12214 Cells w DMSO 55933 Transie siXRC(SMART L-0044 (Dharm was ca Lipofe(On the cells w conflue transfe prepar serum with 3(incuba before ml RPM siRNA, minute tempe then a(plate c

GSE ID	Title	Summary	Overa
			serum- was re supple hours, incuba 48 hou
GSE253354	Distinct pathways for removal of defective RNA polymerase II transcription complexes at a promoter-proximal pause checkpointm [RN21236 dTAG TTchem-seq]	The purpose and function of Integrator and RNA polymerase II (RNAPII) promoter-proximal pausing remain uncertain. Here, we show that when Integrator function is compromised by loss of INTS6, RNAPII interacts increasingly with proteins from alternative pathways for its DNA dissociation, including CUL3-ARMC5 (CRL3ARMC5), which ubiquitylates Ser5-phosphorylated Rpb1 and targets it for degradation. ARMC5-dependent RNAPII ubiquitylation is activated by defects in factors required for correctly regulated promoter-proximal pausing, including Integrator, DSIF and mRNA capping enzyme. This ARMC5 checkpoint curtails an appreciable fraction of RNAPII transcription, with ARMC5 knockout cells producing uncapped, nascent RNA transcripts that fail to mature into stable mRNA.	TTcher 2020) rapid c HEK29 where harbou degror

GSE ID	Title	Summary	Overa
		Concomitant loss of INTS6 and ARMC5 greatly stabilizes RNAPII at the pause and has severe consequences for cell growth and viability. Our data support a model in which CRL3ARMC5 functions alongside Integrator in a checkpoint mechanism that removes faulty RNAPII complexes at promoter-proximal pause sites to safeguard transcription integrity.	
GSE253353	Distinct pathways for removal of defective RNA polymerase II transcription complexes at a promoter-proximal pause checkpoint [RN21114 Ttchemn-Seq]	The purpose and function of Integrator and RNA polymerase II (RNAPII) promoter-proximal pausing remain uncertain. Here, we show that when Integrator function is compromised by loss of INTS6, RNAPII interacts increasingly with proteins from alternative pathways for its DNA dissociation, including CUL3-ARMC5 (CRL3ARMC5), which ubiquitylates Ser5-phosphorylated Rpb1 and targets it for degradation. ARMC5-dependent RNAPII ubiquitylation is activated by defects in factors required for correctly regulated promoter-proximal pausing, including Integrator, DSIF and mRNA capping	TTcher 2020) siRNA- INTS6 cells.

GSE ID	Title	Summary	Overa
		enzyme. This ARMC5 checkpoint curtails an appreciable fraction of RNAPII transcription, with ARMC5 knockout cells producing uncapped, nascent RNA transcripts that fail to mature into stable mRNA. Concomitant loss of INTS6 and ARMC5 greatly stabilizes RNAPII at the pause and has severe consequences for cell growth and viability. Our data support a model in which CRL3ARMC5 functions alongside Integrator in a checkpoint mechanism that removes faulty RNAPII complexes at promoter-proximal pause sites to safeguard transcription integrity.	
GSE307860	Maintenance of gene expression homeostasis by systems-level feedback loops in response to changes in RNA polymerase II dosage [RNAPII_levels_TTseq]	Transcription is regulated by sequence-specific transcription factors and enzymes allowing access to genes in chromatin. However, recent data indicate that the abundance of RNA polymerase II (RNAPII) may itself under certain circumstances represent an additional, crucial determinant of transcription regulation. Here, we titrated the cellular dosage of human	To inve RPB1 a transc RPB1 l degror concer additic expres promo FRT lo of ove hours, metab short p analog Transie

GSE ID	Title	Summary	Overa
		RPB1, the largest RNAPII subunit, using the dTAG system to more generally assess the importance of RNAPII levels. Unexpectedly, cells are extremely sensitive to RPB1 dosage, with a mere 30% reduction sufficient to perturb cell proliferation, cell cycle progression, and global transcription. Using a combination of sequencing and proteomic approaches, we uncover the existence of multiple feedback loops between transcriptional initiation, promotor-proximal pause release, transcript elongation, splicing, and mRNA half-life, which facilitate the maintenance of gene expression homeostasis. Together, our data suggests that gene expression is highly sensitive to RPB1 dosage and that the cellular levels of RNAPII must be tightly regulated.	sequer then u purify labelle resolut transc
GSE169435	The homeotherm-conserved protein CGGBP1 connects heat stress to chromosomal fusions	We report that heat stress induces chromosomal fusion events in cell culture. We describe the findings first using normal primary fibroblasts [Coriell Cell Repository GM02639] and subsequently apply it	Genom MeDIP GM01 descri 32727 on Ion genoty parent

GSE ID	Title	Summary	Overa
		to a system of HEK293T stable lines called CT and KD [described in PMID 31547883]. The stable line CT stands for Control shRNA and KD stands for a knockdown of human CGGBP1 using specific shRNA [Origene Lentiviral system TL313955V]. Our results suggest that as measured by the frequency of chromosomal fusions, the CGGBP1 knockdown in KD, when compared to CT, imitates the effects of heat stress. Further, ChIP-sequencing for TP53BP1 show genome-wide misdirected repairs at repeat flanks in KD as compared to CT. We show that CGGBP1 depletion and heat give rise to chromosomal fusions due to overlapping mechanisms that involve misdirected repair in the flanks of repeats genome-wide.	(GM01 from C also se platfor param- alignm as des 32727
GSE335232	Oxidative stress triggers RNAPII arrest through PARylation and DNA damage [ELCAP-seq]	UV or gamma irradiation, as well as certain chemicals, generate DNA damage that disrupts transcription through a variety of well-characterised mechanisms. In contrast, the transcriptional	ELCAF small F RPB3-I (phosp Ab (un 1mM H recove

GSE ID	Title	Summary	Overa
		response to oxidative stress remains poorly understood. Here, we describe a rapid and widespread shutdown of transcription following oxidative DNA base damage. By monitoring RNAPII occupancy and elongation dynamics, we demonstrate that oxidative stress temporarily halts RNAPII pause release and arrests the progression of elongation complexes within the gene body. We present evidence that this occurs in a unique and transient manner, characterised by abrupt arrest of elongating RNAPII dead in its tracks, followed by rapid transcriptional recovery as DNA lesions are repaired. We find that the restriction of initiation and early elongation complexes is regulated by PARylation, whereas recovery of RNAPII arrested within the gene body requires DNA repair mediated by the base excision repair (BER) and single-strand break repair (SSBR) pathways.	
GSE235181	Promoter-proximal RNA polymerase II termination regulates	Metazoan gene transcription by RNA polymerase (Pol) II is	mNET-seq | T availab

GSE ID	Title	Summary	Overa
	transcription during human cell type transition	regulated in the promoter-proximal region. Pol II can undergo termination in the promoter-proximal region, but whether this can contribute to transcription regulation in cells remains unclear. Here, we extend our previous multi-omics analysis to quantify changes in transcription kinetics during a human cell type transition event. We observe that upregulation of transcription involves an increase in initiation frequency and, at a set of genes, a decrease in promoter-proximal termination. In turn, downregulation of transcription involves a decrease in initiation frequency and an increase in promoter-proximal termination. Thus, promoter-proximal termination of Pol II contributes to the regulation of human gene transcription.	analyze GSM3 SAMN | GSM SAMN | GSM SAMN | GSM SAMN | GSM SAMN | GSM SAMN | GSM SAMN | GSM SAMN | GSM SAMN
GSE71848	Mitotic Transcriptional Clearance of Actively Engaged Pol II Through Transcriptional Elongation Control	Chromatin is highly condensed and transcriptionally repressed during mitosis. Although it is established that some general transcription factors are inactivated by phosphorylation at mitosis,	ChIP-S forms, human cell cy Pol II ir or with flavopi Nascer

GSE ID	Title	Summary	Overa
		many details of mitotic transcriptional repression and its underlying mechanisms are largely unknown. Here, we provide evidence that as cells enter mitosis, genes with transcriptionally engaged RNA Polymerase II (Pol II) can continue transcription until the end of the gene to clear Pol II from mitotic chromatin. Using ChIP-Seq, we find that the transcriptional reinitiation process is globally impaired in early mitosis (prophase/prometaphase), with loss of TFIIB occupancy and nucleosome-free regions at promoters. Pretreatment of nocodazole-arrested mitotic cells with the P-TEFb inhibitor flavopiridol prevents the release of promoter-proximal engaged Pol II. Global nascent RNA sequencing and RNA fluorescence in situ hybridization (FISH) of individual genes demonstrate the existence of transcriptionally engaged Pol II in early mitosis. Chemical and mutational inhibition of P-TEFb in mitosis leads to delays in the progression	asynch mitotic

GSE ID	Title	Summary	Overa
		of cell division. Together, our study reveals a novel mechanism for mitotic transcriptional repression whereby transcriptionally engaged Pol II can progress into productive elongation and finish transcription to allow proper cellular division.	
GSE214557	RNA Polymerase II pausing temporally coordinates cell cycle progression and erythroid differentiation [NET-seq]	The controlled release of promoter-proximal RNA polymerase II (Pol II) pausing into productive elongation is a major step in gene regulation. Yet, functionally analyzing Pol II pausing is difficult because the factors that regulate pause release are generally essential. Here, we identified heterozygous loss-of-function mutations in SUPT5H, encoding SPT5, in individuals with -thalassemia trait unlinked to the -globin locus. Recapitulating the pathogenic mutations in human erythroid cells or acute treatment with transcription inhibitors replicates the patient phenotype of altered globin gene expression. Furthermore, in SUPT5H-edited cells, Pol II pause release was globally disrupted. We observed dynamic patterns of	SUPT5 hemat proger recapit patient imapct erythr RNA-s FACS

GSE ID	Title	Summary	Overa
		pausing at cell cycle and erythroid differentiation genes at the transition from progenitors to precursors during erythropoiesis. At this same transition, in SUPT5H-edited cells, we observed delayed differentiation and altered cell cycle kinetics, as well as a delay in the onset of gene expression programs. Therefore, hindering pause release perturbs cell cycle and delays differentiation at key stages of erythropoiesis, revealing a role for Pol II pausing in the temporal coordination between the cell cycle and differentiation.	
GSE261053	Bypass of blocking lesions by RNAPII reveals key insights into DNA damage resistance [PRO-seq]	Ultraviolet (UV) irradiation or platinum-based drugs generate bulky DNA lesions that impede transcription elongation by RNA Polymerase II (RNAPII). This unscheduled transcription elongation block triggers a stress response, which includes initiating transcription-coupled DNA repair, removing and degrading the halted RNAPII, and a global transcriptional shutdown. However, the molecular and cellular	To cha p300 c transcr damag Sequer conduc p300-c maps t engage paused base-p

GSE ID	Title	Summary	Overa
		consequences of RNAPII successfully bypassing these blocking lesions have remained unknown. Here, we have identified the acetyltransferase p300 as a regulator of the stress response when RNAPII stalls at blocking lesions. Indeed, p300 interacts with the halted RNAPII and controls its removal and degradation in a USP7-dependent mechanism. Notably, in p300-deficient cells, transcription persists despite DNA damage, as RNAPII bypasses blocking lesions, leading to mutant mRNAs and splicing variants accumulation. This persistent transcription activity is particularly detrimental post DNA damage as it results in toxic translation activity and higher DNA damage levels. These observations provide the first evidence that bypass of blocking lesions by RNAPII inhibits DNA repair as well as the cellular stress response. Our findings offer essential insights into the mechanisms of resistance to DNA damage and suggest strategies to	

GSE ID	Title	Summary	Overa
		enhance the efficacy of platinum-based drugs, such as cisplatin, in anti-tumor therapies.	
GSE60358	Mammalian NET-seq reveals genome-wide nascent transcription coupled to RNA processing	We have generated single-nucleotide resolution, nascent transcription profiles from HeLa cells by developing Native Elongation Transcript sequencing technology for mammalian chromatin (mNET-seq). Our extensive data sets provide a substantial resource to study mammalian nascent transcript profiles. We reveal unanticipated phosphorylation states for RNA polymerase II C-terminal domain (Pol II CTD) at both gene ends. We also observe that following 5 splice site cleavage by the spliceosome, upstream exon transcripts are tethered to Pol II CTD phosphorylated on the serine 5 position (S5P) which is accumulated over downstream exons. We further show that depletion of termination factors substantially reduces Pol II pausing at gene ends leading to termination defects. Remarkably termination factors play an additional	To mo the ma and its differe states, metho comple sequer bound mNET- using t CMA6 CMA6 for un Ser2 p phospl isoforr anothe evalua transc factors produc and cc ChrRN on Hel siRNA CstF64 and th compa HeLa t for Luc the sa

GSE ID	Title	Summary	Overa
		promoter role by restricting non-productive RNA synthesis and redistributing Pol II CTD S2P to promoters. These data demonstrate that CTD phosphorylation is more dynamic and variably distributed across mammalian transcription units than previously envisaged.	
GSE109652	Transcription elongation rate affects nascent histone pre-mRNA folding and 3 end processing	We report that slow transcription by mutant RNA pol II caused accumulation of polyadenylated histone mRNAs that extend past the stem loop processing site. UV irradiation, which decelerates pol II elongation, also induced long poly(A)+ histone transcripts. Inhibition of 3 processing by slow pol II correlates with failure to recruit SLBP to histone genes. Chemical probing of nascent RNA structure showed that the stem loop fails to fold in transcripts made by slow pol II, thereby explaining the absence of SLBP and failure to process 3 ends. These results show that regulation of transcription speed can modulate pre-mRNA processing by changing nascent RNA	The ef elonga Bindin Negati (Nelf) a polyad (CSTF ChIP-s HEK29 induci amanit the RN subuni rates. RNA is targete with DI employ nascer replica loop. .

GSE ID	Title	Summary	Overa
		structure, and suggest a mechanism by which alternative processing could be controlled	
GSE97527	PAF1 regulation of promoter-proximal pause release via enhancer activation	Gene expression in metazoans is regulated by RNA Polymerase II (Pol II) promoter-proximal pausing and its release. Previously, we identified that Pol II-associated factor 1 (PAF1) modulates the release of paused Pol II into productive elongation. Here, we find that PAF1 occupies transcriptional enhancers and restrains hyperactivation of a subset of these enhancers. Enhancer activation as the result of Paf1 loss releases Pol II from paused promoters of nearby PAF1 target genes. Knockout of PAF1-regulated enhancers attenuates the release of paused Pol II on PAF1 target genes without major interference in the establishment of pausing at their cognate promoters. Thus, a subset of enhancers can primarily modulate gene expression by controlling the release of paused Pol II in a PAF1-dependent manner.	ChIP-s of RNA lines ta scramb in cells using t degror

GSE ID	Title	Summary	Overa
GSE254796	Disruption of RNAPII transcription elongation links Oncogenic splicing factor mutations to replciation stress and targetable alterations in chromatin landscape [NTD-Seq]	To investigate changes in RNAPII distribution in response to SF3B1 mutations, we peformed ChIP-seq using antibody against N-terminal domain of RNAPII (Pan-Pol2) in K562 cells | We then performed coverage plot analysis using data obtained from ChIP-seq to investigate RNAPII distribution changes	ChIP-s in K56 conditi
GSE136024	Genome-wide view of the impact of Spt5-Pol II inhibitors (SPIs) on transcription [GRO-Seq]	We identified the first Spt5-Pol II inhibitors (SPIs). SPIs faithfully reproduced Spt5 knockdown effects on proximal-promoter-pausing, NF-B activation and the expanded-repeat huntingtin gene in neuronal cells. Using SPIs we identified Spt5 target genes that responded with profoundly diverse kinetics and a novel regulatory element of proximal-promoter-pausing. To obtain a genome-wide view of the impact of SPIs on transcription, we performed Global run-on sequencing (GRO-seq) using nuclei of HeLa cells treated with either DMSO (control) or SPI-21.	Nuclei cells p DMSO (50M)

GSE ID	Title	Summary	Overa
GSE231486	Genome-wide Transcription Factor binding maps reveal cell-specific changes in the regulatory architecture of human HSPC	This SuperSeries is composed of the SubSeries listed below.	Refer t
GSE269116	Notch induces transcription by stimulating release of paused RNA Pol II without increasing chromatin accessibility [TT-seq]	Notch proteins undergo ligand-induced proteolysis to release a nuclear effector that influences a wide range of cell fate decisions by regulating gene activity. However, despite years of study, how Notch induces the transcription of its target genes remains unclear. Here, we comprehensively examined the response to human Notch1 across a time course of activation, using genomic assays of nascent RNA and chromatin accessibility. These data revealed that Notch induces target gene transcription primarily by releasing paused RNA polymerase II (RNAPII), in contrast to prevailing models suggesting that Notch acts by promoting chromatin accessibility. Indeed, we found that open chromatin is established at Notch-responsive regulatory elements prior to Notch	To det induce SC2 ce inhibiti expres respor induce washo at 1 ar activat after n harves prepar contro harves For ea cells w SWI/SI BRM0 the tim washo conditi of the

GSE ID	Title	Summary	Overa
		signaling, through SWI/SNF-mediated remodeling. Notch activation, however, elicited no further chromatin opening at these loci. Together, these studies reveal that the nuclear response to Notch signaling is dictated by the pre-existing chromatin state and RNAPII distribution at the time of signal activation.	
GSE132395	Elongation Factor TFIIS Prevents Transcription Stress and R-Loop Accumulation to Maintain Genome Stability [TT-seq]	Connections between RNA polymerase II (RNAPII) transcription stress, R-loops, and genome instability have been established however, the underlying mechanisms remain poorly understood. Here we used a mutant version of elongation factor TFIIS (TFIISmut) to specifically induce increased levels of RNAPII pausing, arrest, and/or backtracking in human cells. TFIISmut expression results in slower elongation rates, relative depletion of polymerases from the end of genes, and increased levels of stopped RNAPII. It affects mRNA splicing and termination as well. Remarkably, however, TFIISmut expression also	To stu and its we use which under promo form c elonga for stir cleava cells w preser over-e prior t

GSE ID	Title	Summary	Overa
		dramatically increases R-loops, which may form at the anterior end of backtracked RNAPII and trigger genome instability, including DNA strand breaks. These results shed new light on the relationship between transcription stress and R-loops, and suggest that different classes of R-loops exist, potentially with distinct consequences for genome instability.	
GSE335234	Oxidative stress triggers RNAPII arrest through PARylation and DNA damage [TTchem-seq]	UV or gamma irradiation, as well as certain chemicals, generate DNA damage that disrupts transcription through a variety of well-characterised mechanisms. In contrast, the transcriptional response to oxidative stress remains poorly understood. Here, we describe a rapid and widespread shutdown of transcription following oxidative DNA base damage. By monitoring RNAPII occupancy and elongation dynamics, we demonstrate that oxidative stress temporarily halts RNAPII pause release and arrests the progression of elongation complexes within the gene body. We	TTcher 15min recove various Olapar Knock

GSE ID	Title	Summary	Overa
		present evidence that this occurs in a unique and transient manner, characterised by abrupt arrest of elongating RNAPII dead in its tracks, followed by rapid transcriptional recovery as DNA lesions are repaired. We find that the restriction of initiation and early elongation complexes is regulated by PARylation, whereas recovery of RNAPII arrested within the gene body requires DNA repair mediated by the base excision repair (BER) and single-strand break repair (SSBR) pathways.	
GSE195851	Integrative nascent RNA methods to reveal cell-type specific transcription programs in peripheral blood and its derivative cells	Nascent RNA sequencing is a powerful method to measure transcription with high resolution, sensitivity, and directional information, which gives distinctive information about transcription from other methods such as chromatin immunoprecipitation or mRNA sequencing. We present an integrated package of nascent RNA-seq methods - ultrafast Precision Run On (uPRO) combined with computational procedures to discover cell type	Nascer periph cell lin

GSE ID	Title	Summary	Overa
		specific enhancers, promoters, and transcription factor networks. uPRO is composed of adaptor ligation and reverse transcription reactions, which is reduced to a one-day procedure and makes nascent RNA-seq more feasible and flexible for a widespread use. We generated genome-wide profiles of nascent transcription in human blood derived cell lines and clinical samples of ~1 ml of untreated whole blood. We integrated these data into deep learning and hierarchical network analysis to detect enhancers, promoters, and co-expression networks to define cell-type specific transcription programs. We found conservation of position but variation of expression in cell type specific enhancers and transcription start sites. Transcription factors (TFs) such as TCF-3 and OCT1 were pivotally associated with TF-enhancer-gene networks across cell types. Intriguingly, we also discovered that TFs related to cell stress and inflammation - such as	

GSE ID	Title	Summary	Overa
		SRF, ATF, CHOP, and NF-kB - are associated with inter-individual variation of leukocyte transcription in whole blood. Our integration of experimental and computational nascent RNA methods will provide an efficient strategy to identify specific transcriptional programs, both in cell-type and patient/disease-associated, with minimal sample requirements.	
GSE136025	Genome-wide view of the impact of Spt5-Pol II inhibitors (SPIs) on transcription [4sU-seq]	We identified the first Spt5-Pol II inhibitors (SPIs). SPIs faithfully reproduced Spt5 knockdown effects on proximal-promoter-pausing, NF-B activation and the expanded-repeat huntingtin gene in neuronal cells. Using SPIs we identified Spt5 target genes that responded with profoundly diverse kinetics and a novel regulatory element of proximal-promoter-pausing. To validate that the effects of SPIs are at the transcriptional level, cellular RNA was metabolically labeled with 4-thio-uridine (4sU) for 2 hours in the presence of DMSO or SPI-21 in the presence or absence of	Cellula labelec (4sU) 1 preser in the TNF.

GSE ID	Title	Summary	Overa
		TNF. Newly synthesized labeled RNA was then purified and subjected to RNA-seq.	
GSE96056	CDK9-dependent RNA polymerase II pausing controls transcription initiation	Gene transcription can be activated by decreasing the duration of RNA polymerase II pausing in the promoter-proximal region, but how this is achieved remains unclear. Here we use a multi-omics approach to show that the duration of polymerase pausing generally limits the frequency of transcription initiation in human cells (pause-initiation limit). We further engineer a human cell line to allow for specific and rapid inhibition of the P-TEFb kinase CDK9, which is implicated in polymerase pause release. We show that CDK9 activity decreases the pause duration but also increases the initiation frequency. This shows that CDK9 stimulates release of paused polymerase and activates transcription by increasing the number of transcribing polymerases and thus the amount of mRNA synthesized per time. CDK9 activity is also associated with long-range chromatin interactions,	We pe reactic BB det immun biologi Raji B cells tr DMSO PP1 (C perfor includi biologi Raji B or CDk 15 mir 5 M 1- inhibit of 4sU lympho were e CRISPI contai mutati gene l

GSE ID	Title	Summary	Overa
		suggesting that enhancers can influence the pause-initiation limit to regulate transcription.	
GSE197247	Effect of Heat stress on m6A modification in CaSki cell	N6-methyladenosine (m6A) occurs extensively on multiple RNA species and is the most abundant internal modification of eukaryotic messenger RNA (mRNA). RNA m6A methylation is a dynamic and reversible process subjected to regulation by three categories of m6A modifier proteins, including m6A methyltransferases/writers (e.g., METTL3/14 methyltransferase complex and METTL16), m6A demethylases/erasers (e.g., FTO and ALKBH5) and m6A binding proteins/readers (e.g., YTH family proteins and IGF2BPs). The dynamic and adjustable nature makes m6A a key regulator in RNA metabolism, and the fate of m6A-modified RNA is modulated by m6A readers. Previous studies have demonstrated that fever can benefit survival of cancer patients, and currently hyperthermia (elevating body	CaSki heat tr before extrac sequer

GSE ID	Title	Summary	Overa
		temperature beyond normal) is either administered alone or in combination with radiotherapy or chemotherapy to treat various neoplasms with unclear mechanisms. Accruing evidence has revealed favorable effects of fever/hyperthermia in cancer treatment through affecting the properties of oncogenic proteins (e.g., stability, posttranslational modification, ability to interact with other molecules). Accordingly, we sought to investigate whether and how heat stress affects cancer development by m6A modification.	
GSE156339	Stress-induced RNAchromatin interactions promote endothelial dysfunction (iMARGI replicate 2)	Chromatin-associated RNA (caRNA) has been proposed as a type of epigenomic modifier. Here, we test whether environmental stress can induce cellular dysfunction through modulating RNA-chromatin interactions. We induce endothelial cell (EC) dysfunction with high glucose and TNF (H+T), that mimic the common stress in diabetes mellitus. We characterize the H+T-induced changes in gene expression by single cell	Map R HUVE iMARG sample treatm treatm to mes

GSE ID	Title	Summary	Overa
		(sc)RNA-seq, DNA interactions by Hi-C, and RNA-chromatin interactions by iMARGI. H+T induce inter-chromosomal RNA-chromatin interactions, particularly among the super enhancers. To test the causal relationship between H+T-induced RNA-chromatin interactions and the expression of EC dysfunction-related genes, we suppress the LINC00607 RNA. This suppression attenuates the expression of SERPINE1, a critical pro-inflammatory and pro-fibrotic gene. Furthermore, the changes of the co-expression gene network between diabetic and healthy donor-derived ECs corroborate the H+T-induced RNA-chromatin interactions. Taken together, caRNA-mediated dysregulation of gene expression modulates EC dysfunction, a crucial mechanism underlying numerous diseases.	

GSE ID	Title	Summary	Overa
GSE316831	Transcriptomic impacts of TFIIB D207G editing in Jurkats during 2 h TRAIL treatment	Cellular stressors often cause widespread repression of RNA polymerase II (RNAP II) activity, which is thought to facilitate a focused transcriptional output towards stress resolution. In many cases, however, the underlying regulatory mechanisms remain unknown. Here, we demonstrate that stress-induced downregulation of the general transcription factor TFIIB tempers expression of specific stimulus response genes. Following a variety of stressors, TFIIB is proteolytically cleaved between its cyclin folds at conserved aspartic acid residue D207 by caspases- 3 and 7. Cleavage in this portion of the protein significantly reduces the ability of TFIIB to form a TBP-TFIIB-DNA promoter complex in vitro. Using both overexpression and endogenous base-editing, we find that B and T cells that are unable to cleave TFIIB upregulate expression of a select gene set during apoptosis. These TFIIB-sensitive genes are primarily short, stimulus-responsive and	Using sensiti two dif the TF validat D207G sgRNA sequer dinucle made i (Contr popula at eacl validat sequer popula 10 day base e sgRNA SLAM- stabilit experi impact paralle Cells w SUPEF and 4s RNA) f harves by RN library sequer gene e

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		proto-oncogenic loci, and cleavage of TFIIB temporally restricts their expression. Failure to cleave TFIIB during stress leads to aberrant lymphocyte proliferation during chemical perturbation. Hence, caspase targeting of TFIIB destabilizes transcription to tune gene expression, allowing for proper stress resolution.	
GSE254793	Disruption of RNAPII transcription elongation links Oncogenic SFB31 mutations to replciation stress and targetable alterations in chromatin landscape [mNET-Seq]	To investigate changes in RNAPII distribution in response to SF3B1 mutations, we peformed mNET-seq using antibody against either pan Pol II CTD or Ser2P Pol II CTD in K562 cells (wild type and SF3B1K700E mutant) at 4 days after mutant allele expression | We then performed coverage plot analysis using data obtained from mNET-seq to investigate RNAPII distribution changes. Drosophila spike-in was included for the pan-Pol II CTD studies.	mNET- K562 (SF3B1 4 days inducti
GSE269115	Notch induces transcription by stimulating release of paused RNA Pol II without increasing chromatin	Notch proteins undergo ligand-induced proteolysis to release a nuclear effector that influences a wide range of cell fate decisions by regulating	To det signali of RNA signali cells b were h

GSE ID	Title	Summary	Overa
	accessibility [PRO-seq]	gene activity. However, despite years of study, how Notch induces the transcription of its target genes remains unclear. Here, we comprehensively examined the response to human Notch1 across a time course of activation, using genomic assays of nascent RNA and chromatin accessibility. These data revealed that Notch induces target gene transcription primarily by releasing paused RNA polymerase II (RNAPII), in contrast to prevailing models suggesting that Notch acts by promoting chromatin accessibility. Indeed, we found that open chromatin is established at Notch-responsive regulatory elements prior to Notch signaling, through SWI/SNF-mediated remodeling. Notch activation, however, elicited no further chromatin opening at these loci. Together, these studies reveal that the nuclear response to Notch signaling is dictated by the pre-existing chromatin state and RNAPII distribution at the time of signal activation.	45m, 1 post N harves prepar conditi immed washo PRO-se There with tw experi the ref

GSE ID	Title	Summary	Overa
GSE214556	RNA Polymerase II pausing temporally coordinates cell cycle progression and erythroid differentiation [CRISPR]	The controlled release of promoter-proximal RNA polymerase II (Pol II) pausing into productive elongation is a major step in gene regulation. Yet, functionally analyzing Pol II pausing is difficult because the factors that regulate pause release are generally essential. Here, we identified heterozygous loss-of-function mutations in SUPT5H, encoding SPT5, in individuals with -thalassemia trait unlinked to the -globin locus. Recapitulating the pathogenic mutations in human erythroid cells or acute treatment with transcription inhibitors replicates the patient phenotype of altered globin gene expression. Furthermore, in SUPT5H-edited cells, Pol II pause release was globally disrupted. We observed dynamic patterns of pausing at cell cycle and erythroid differentiation genes at the transition from progenitors to precursors during erythropoiesis. At this same transition, in SUPT5H-edited cells, we observed delayed differentiation and altered	SUPT5 hemat proger recapit patient imapct erythr RNA-s FACS

GSE ID	Title	Summary	Overa
		cell cycle kinetics, as well as a delay in the onset of gene expression programs. Therefore, hindering pause release perturbs cell cycle and delays differentiation at key stages of erythropoiesis, revealing a role for Pol II pausing in the temporal coordination between the cell cycle and differentiation.	
GSE307769	Maintenance of gene expression homeostasis by systems-level feedback loops in response to changes in RNA polymerase II dosage	Transcription is regulated by sequence-specific transcription factors and enzymes allowing access to genes in chromatin. However, recent data indicate that the abundance of RNA polymerase II (RNAPII) may itself under certain circumstances represent an additional, crucial determinant of transcription regulation. Here, we titrated the cellular dosage of human RPB1, the largest RNAPII subunit, using the dTAG system to more generally assess the importance of RNAPII levels. Unexpectedly, cells are extremely sensitive to RPB1 dosage, with a mere 30% reduction sufficient to perturb cell proliferation, cell cycle	To inve RPB1 proxim modul treatin with 4 dTAGv were h perfor run-on extrac (PRO-s the ge of trar RNA p nucleo

GSE ID	Title	Summary	Overa
		progression, and global transcription. Using a combination of sequencing and proteomic approaches, we uncover the existence of multiple feedback loops between transcriptional initiation, promotor-proximal pause release, transcript elongation, splicing, and mRNA half-life, which facilitate the maintenance of gene expression homeostasis. Together, our data suggests that gene expression is highly sensitive to RPB1 dosage and that the cellular levels of RNAPII must be tightly regulated.	
GSE302707	Transcriptomic impacts of TFIIB D207G editing in Jurkats during 6 h TRAIL treatment	Cellular stressors often cause widespread repression of RNA polymerase II (RNAP II) activity, which is thought to facilitate a focused transcriptional output towards stress resolution. In many cases, however, the underlying regulatory mechanisms remain unknown. Here, we demonstrate that stress-induced downregulation of the general transcription factor TFIIB tempers expression of specific stimulus response genes. Following a variety of	Using sensiti two dif the TF validat D207C sgRNA contro was m (Contr popula at eacl validat sequer popula 10 day base e sgRNA SLAM-

GSE ID	Title	Summary	Overa
		stressors, TFIIB is proteolytically cleaved between its cyclin folds at conserved aspartic acid residue D207 by caspases- 3 and 7. Cleavage in this portion of the protein significantly reduces the ability of TFIIB to form a TBP-TFIIB-DNA promoter complex in vitro. Using both overexpression and endogenous base-editing, we find that B and T cells that are unable to cleave TFIIB upregulate expression of a select gene set during apoptosis. These TFIIB-sensitive genes are primarily short, stimulus-responsive and proto-oncogenic loci, and cleavage of TFIIB temporally restricts their expression. Failure to cleave TFIIB during stress leads to aberrant lymphocyte proliferation during chemical perturbation. Hence, caspase targeting of TFIIB destabilizes transcription to tune gene expression, allowing for proper stress resolution.	stabilit experi impact paralle Cells w SUPEF 6 h an label n hours was fo extrac prepar and di expres
GSE125535	The Human Integrator Complex Facilitates Transcriptional Elongation by	Transcription by RNA polymerase II (RNAPII) is pervasive in the human genome. However, the mechanisms controlling	Pro-se contro shGFP, INTS1

GSE ID	Title	Summary	Overa
	Endonucleolytic Cleavage of Nascent Transcripts	transcription at promoters and enhancers remain enigmatic. Here, we demonstrate that Integrator subunit 11 (INTS11), the catalytic subunit of the Integrator complex, regulates transcription at these loci through its endonuclease activity. Promoters of genes require INTS11 to cleave nascent transcripts associated with paused RNAPII and induce their premature termination in the proximity of the +1 nucleosome. The turnover of RNAPII permits the subsequent recruitment of an elongation-competent RNAPII complex, leading to productive elongation. In contrast, enhancers require INTS11 catalysis not to evict paused RNAPII but rather to terminate enhancer RNA transcription beyond the +1 nucleosome. These findings are supported by the differential occupancy of negative elongation factor (NELF), SPT5, and tyrosine-1-phosphorylated RNAPII. This study elucidates the role of Integrator in mediating transcriptional elongation at human promoters	

GSE ID	Title	Summary	Overa
		through the endonucleolytic cleavage of nascent transcripts and the dynamic turnover of RNAPII.	
GSE161671	The Nuclear Transcription factor TAF7 is a Cytoplasmic RNA Chaperone	This SuperSeries is composed of the SubSeries listed below.	Refer t
GSE147055	The 7SK/P-TEFb snRNP controls ultraviolet radiation-induced transcriptional reprogramming	Releasing promoter-proximally-paused RNA polymerase II (RNAPII) by the positive transcription elongation factor b (P-TEFb) is a central regulatory step of eukaryotic mRNA synthesis. The nuclear activity of P-TEFb is controlled mainly by the 7SK small nuclear RNP (snRNP) which sequesters active P-TEFb into inactive 7SK/P-TEFb snRNP. Here we demonstrate that under normal culture conditions, the lack of 7SK snRNP has only a minor impact on global RNAPII transcription and promoter-proximal pausing without detectable consequences on cell growth and proliferation. However, upon ultraviolet (UV) light-induced DNA damage, cells lacking 7SK have a defective	ChIRP-HAP1-mNET-HAP1 -/+ 1h,

GSE ID	Title	Summary	Overa
		transcriptional response and highly reduced viability. Both UV-induced release of lesion-scanning polymerases and activation of key early responsive genes are compromised in the absence of 7SK. We show that proper induction of 7SK-dependent UV-responsive genes, namely efficient release of paused RNAPII and recruitment of key elongation factors, requires P-TEFb activity directly mobilized from the nucleoplasmic 7SK/P-TEFb snRNP.	
GSE139464	Selective Mediator-dependence of cell type-specifying transcription [4C-seq]	The Mediator complex directs signals from DNA-binding transcription factors to RNA polymerase (Pol) II. Despite this pivotal position, mechanistic understanding of Mediator in human cells remains incomplete. Here, we quantified Mediator-controlled Pol II kinetics by coupling rapid subunit degradation with orthogonal experimental readouts. Consistent with a model of condensate-driven transcription initiation, large clusters of hypo-phosphorylated Pol II rapidly disassembled upon	4C-sec degrac in KBM

GSE ID	Title	Summary	Overa
		Mediator degradation. This was accompanied by a selective and pronounced disruption of cell type-specifying transcriptional circuits, whose constituent genes featured exceptionally high rates of Pol II turnover. Notably, transcriptional output of most other genes was largely unaffected by acute Mediator ablation. Maintenance of transcriptional activity at these genes was linked to an unexpected, CDK9-dependent compensatory feedback loop that elevated Pol II pause release rates genome-wide. Collectively, our work positions human Mediator as a globally acting coactivator that selectively safeguards the functionality of cell type-specifying transcriptional networks.	
GSE251941	Distinct pathways for removal of defective RNA polymerase II transcription complexes at a promoter-proximal pause checkpoint	The Integrator complex interacts with RNA polymerase II (RNAPII) at promoter-proximal pause sites but its function there remains uncertain. Here, we show that when Integrator function is compromised by loss of INTS6, RNAPII interacts increasingly with proteins	To inve INTS6 seq wa knock

GSE ID	Title	Summary	Overa
		that mediate its dissociation from DNA, including CUL3-ARMC5, which ubiquitylates Ser5-phosphorylated Rpb1 and targets it for degradation. ARMC5-dependent RNAPII ubiquitylation is activated by defects in factors required for correctly regulated promoter-proximal pausing, including Integrator, DSIF and mRNA capping enzyme. This ARMC5 checkpoint curtails an appreciable fraction of RNAPII transcription, with ARMC5 knockout cells producing uncapped, nascent RNA transcripts that fail to mature into stable mRNA. Concomitant loss of INTS6 and ARMC5 greatly stabilizes RNAPII at the pause and has severe consequences for cell growth and viability. Our data support a model in which CRL3ARMC5 functions alongside Integrator in a checkpoint mechanism that removes faulty RNAPII complexes at promoter-proximal pause sites to safeguard transcription integrity.	

GSE ID	Title	Summary	Overa
GSE134102	NET-seq analysis of pSer2-Pol II profile in shCTRL, shMETTL3 and shDDX21 HeLa cells	The chromatin fraction was isolated from 5x107 HeLa cells. Chromatin was digested in 100 mL of MNase (40 units/ mL) reaction buffer for 3 min at 37 C in a thermomixer (1,400 rpm). After addition of 10 L EGTA (25mM) to inactivate MNase, soluble digested chromatin was collected by 13,000 rpm centrifuge for 5 min. 10 mg of Pol II antibody was conjugated to magnetic protein A beads. The supernatant was diluted with 400 mL of NET-2 buffer and pSer2-Pol II antibody-conjugated beads were added. Incubated the IP reaction at 4 C for 1 hr. The beads were washed with 1 mL of NET-2 buffer six times and 100 mL of 1xPNKT (1xPNK buffer and 0.05% Triton X-100) buffer once. Washed beads were incubated in 50 mL PNK reaction mix (1xPNKT, 1 mM ATP and 0.05 U/ml T4 PNK (NEB)) in Thermomixer at 37 C, 1,400 rpm for 6 min. Beads were washed with 1 mL of NET-2 buffer once and RNA was extracted with Trizol reagent. RNA was suspended in urea	Examir profile

GSE ID	Title	Summary	Overa
		Dye (7M Urea, 1xTBE, 0.1% BPB and 0.1% XC) and resolved on 6% TBU gel (Invitrogen) at 200 V for 5 min. Gels with 30-160 nt RNAs were collected. 0.2 mL tube was prepared with holes made with 25G needle and placed in a 1.5 mL tube. Gel fragments were placed in the layered tube and broken down by centrifugation at 12,000 rpm for 1 min. The small RNAs were eluted from gel using RNA elution buffer (1 M NaOAc and 1 mM EDTA) at 25 C for 1 hr in Thermomixer (900 rpm). Eluted RNA was puried with SpinX column (Coster) with 2 glass lters (Millipore) and the ow-through RNA was ethanol precipitated.	
GSE270292	An accessible and convenient assay to profile single-cell newly synthesized transcriptome reveals transcription factors and regulon dynamics during early-stage T-cell activation	Sequencing newly synthesized transcriptome in addition to the regular transcriptome in single cells enable the study of gene expression temporal dynamics during rapid chromatin and gene regulation processes. However, current single-cell newly synthesized transcriptome assays require in-house technology expertise to	eHAP subjec 10xGe v3.1 ki cultiva PMA/I DMSO NOTE- nave T antiCD stimula seq pr experi NIH3T

GSE ID	Title	Summary	Overa
		achieve a high cellular throughput, preventing their widespread application. Here, we develop NOTE-seq to simultaneously profile regular and newly synthesized transcriptome in single cells. NOTE-seq combines 4-thiouridine labeling of newly synthesized RNA, thiol-alkylation-based chemical conversion, and the streamlined 10X Genomics platform, offering a high cellular throughput accessible to and convenient for regular biology labs without single-cell technology expertise. Using NOTE-seq, we investigate the temporal dynamics of gene expression during early-stage T-cell activation in human Jurkat T cells and mouse nave T cells, characterize transcription factors and regulons, and discover Fli-1 as a master transcription factor for gene regulation upon T-cell activation. Interestingly, Fli-1 level in T cells is sensitive to the treatment of camptothecin, a topoisomerase inhibitor	1:1 rat NOTE- to-C c

GSE ID	Title	Summary	Overa
		used in cancer chemotherapy, indicating its potential complication on the immune system.	
GSE292670	Analysis of nascent transcription by EU-seq in HUDEP-2 cells following CTCF degredation	CTCF plays a crucial role in looping and insulation, yet its role in BCL11A regulation remains unclear. Here, we utilize EU-seq to measure nascent transcription in HUDEP-2 cells with CTCF-AID.	EU-sec cells w for 0h,
GSE287021	Capped small RNA sequencing (csRNA-seq) for genome-wide mapping of active transcription start sites and enhancer RNAs from total RNA	Profiling ongoing transcription reveals the complete active transcriptome, including unstable enhancer RNAs, and provides a dynamic view of transcription dynamics, enabling the study of underlying gene regulatory mechanisms. Here, we describe a detailed protocol for capped- small RNA-sequencing (csRNA-seq), a method that uses total RNA to capture actively initiating or nascent RNA polymerase II transcripts, including unstable RNAs such as enhancer RNAs, and their transcription start sites (TSSs). csRNA-seq offers several advantages: (1) the decoupling of sample collection and processing; (2) compatibility with	The pr RNA is enrichi genera sequer to com input s nascer

GSE ID	Title	Summary	Overa
		diverse sample types, including fresh, frozen, and fixed tissues, as well as various eukaryotic organisms; and (3) scalability. Purified RNA is further non-infectious, can be isolated from inactivated samplesincluding clinical specimens or pathogenic samplestransported, and studied under standard laboratory conditions. This accessible protocol empowers researchers with minimal experience in nascent transcriptomics to study gene regulation and transcription dynamics.	
GSE219965	PRO-cap from HCT116 (ENCSR990FJB)	A run-on assay that precisely maps the transcription start sites genome-wide with high sensitivity and specificity in CDK7_U. | For data usage terms and conditions, please refer to [url] and [url]	[url]
GSE161792	Integrated multiomic analysis identifies functional epigenetic signatures regulating nucleotide excision repair (meRIP-seq)	Transcription-coupled nucleotide excision repair (TC-NER) is promptly triggered by 'local' elongating RNAP II molecules encountering DNA adducts such as cyclobutane pyrimidine dimers (CPD) induced by ultraviolet light and speeds	MRC5 untrea with 1 irradia recove 24 hou replica each s

GSE ID	Title	Summary	Overa
		up removing and repair in transcribed loci. Genome-wide measurements have revealed a dramatic shutdown of global transcription with a few exceptions of individual genes that are consequently highly upregulated by DNA damage. Deciphering the underlying mechanisms of the multifaceted transcriptional response always comprises one of the most fascinating frontiers in DNA repair research. Recent reports have revealed that persistent RNAP II initiation at the transcription start site of active regulatory regions with the continuous synthesis of start-RNAs guarantee sufficient scanning and repair of the whole transcribed genome, although transcription elongation drastically slows down shortly after genotoxic stress. Furthermore, a recent study found that RNA is extremely rapidly m6A methylated by METTL3 in response to UV irradiation which is crucial for recruitment of Pol k to DNA damage sites	

GSE ID	Title	Summary	Overa
		and subsequently mediates TC-NER. Together, these findings substantiate the molecular processes underlying the transcription-coordinated cellular response upon genotoxic stress might be more complex than previously believed. Unfortunately, all these findings merely focused on the early phase of recovery (within 4 hr after DNA damage). It remains to be determined how accessible chromatin reconfigures and regulates transcriptional programs in UV-induced DNA damage repair. It is also unclear whether the dynamics of gene expression are associated with other epigenomic reconstruction events such as global DNA methylation and demethylation. Moreover, the roles of internal messenger RNA modifications like N6-methyladenosine in TC-NER are poorly understood. Our integrative analyses provide a comprehensive view of the spatio-temporal chromatin configuration and	

GSE ID	Title	Summary	Overa
		epigenetic characterizations that accompanies TC-NER.	
GSE225999	Disruption of RNAPII transcription elongation links Oncogenic splicing factor mutations to replciation stress and targetable alterations in chromatin landscape [GRO-seq]	To investigate redistribution of RNAPII in response to the SF3B1 mutation and to study its effects on nascent gene expression, we peformed fastGRO-seq (variant of GRO-seq using 4-thio-UTP based nuclear run-on) in K562 cells (wild type and SF3B1K700E mutant) at 4 days after mutant allele expression | We then performed metaplot-based and gene expression analysis using data obtained from fastGRO-seq	Nuclea (wild ty mutan doxycy
GSE135355	Stress-induced RNAchromatin interactions promote endothelial dysfunction (iMARGI)	Chromatin-associated RNA (caRNA) has been proposed as a type of epigenomic modifier. Here, we test whether environmental stress can induce cellular dysfunction through modulating RNA-chromatin interactions. We induce endothelial cell (EC) dysfunction with high glucose and TNF (H+T), that mimic the common stress in diabetes mellitus. We characterize the H+T-induced changes in gene expression by single cell (sc)RNA-seq, DNA	Map R HUVE(iMARG sample treatm days o endoth transit

GSE ID	Title	Summary	Overa
		interactions by Hi-C, and RNA-chromatin interactions by iMARGI. H+T induce inter-chromosomal RNA-chromatin interactions, particularly among the super enhancers. To test the causal relationship between H+T-induced RNA-chromatin interactions and the expression of EC dysfunction-related genes, we suppress the LINC00607 RNA. This suppression attenuates the expression of SERPINE1, a critical pro-inflammatory and pro-fibrotic gene. Furthermore, the changes of the co-expression gene network between diabetic and healthy donor-derived ECs corroborate the H+T-induced RNA-chromatin interactions. Taken together, caRNA-mediated dysregulation of gene expression modulates EC dysfunction, a crucial mechanism underlying numerous diseases.	
GSE133156	CDK12 globally stimulates RNA polymerase II transcription elongation and carboxyl-terminal	Cyclin-dependent kinase 12 (CDK12) phosphorylates the carboxyl-terminal domain (CTD) of RNA polymerase II (pol II) but its roles in	TT-seq cells at 15 mir with 1-

GSE ID	Title	Summary	Overa
	domain phosphorylation	transcription beyond the expression of DNA damage response genes remain unclear. Here, we have used TT-seq and mNET-seq to monitor the direct effects of rapid CDK12 inhibition on transcription activity and CTD phosphorylation in human cells. CDK12 inhibition causes a genome-wide defect in transcription elongation and a global reduction of CTD Ser2 and Ser5 phosphorylation. The elongation defect is explained by the loss of the elongation factors LEO1 and CDC73, part of PAF1 complex, and SPT6 from the newly-elongating pol II. Our results indicate that CDK12 is a general activator of pol II transcription elongation and indicate that it targets both Ser2 and Ser5 residues of the pol II CTD.	
GSE141370	DRB-PRO-Seq for measuring Pol II transcriptional elongation rate alteration up on CDK12 and CDK13 inhibition by THZ531	We performed PRO-Seq to measure the Pol II elongation rate by measruing the distance of Pol II travels at viarous time points after relased from DRB induced pausing. We compared DMSO/DRB or THZ531/DRB co-treated	Cells w treatm DRB a THZ53 hours, out to for 12 and 45 out, 0 also ha

GSE ID	Title	Summary	Overa
		sample to assess the effect of CDK12 and CDK13 inhibtion by THZ531 on transcriptional elongation rate.	covela CDK13 effect inhibti DRB b sample DMSO 200nM DRB.
GSE116472	Single-molecule nascent RNA sequencing reveals regulatory domain architecture ant promoters and enhancers	CoPRO adapts PRO-cap for paired end sequencing, and includes a total of three different libraries that were enriched for either capped nascent RNAs only, uncapped nascent RNAs only, or both. With paired end sequencing, each read tells us where an RNA polymerase molecule initiated, and then where its active site is located. Comparison of the libraries for different capping states allows us identify the precise location of pausing, and of where nascent RNAs become capped across the tens of thousands of initiation sites that we detect. The paired nature of the data enabled identification of sites of transcription initiation with unprecedented precision: we could use the pattern of pausing and elongation as a sensitive way of	CoPRC nascer Uncap only, o Uncap were p cells, e duplica Cappe were p with ar heat sh

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		calling real initiation sites, and filter out termination by comparing capped and uncapped treatments. Because CoPRO maps nascent RNA, it is not affected by the post-transcriptional stability of the initiation and pausing events detected. Thus, it puts non-coding transcription (such as eRNAs and upstream divergent RNAs) on an equal footing with longer lived RNAs like mRNA and lncRNAs. With our comprehensive maps of transcription initiation, we are able to compare the architecture of initiation sites with other features of the genome. For this purpose, we chose human K562 cells for direct comparison with dozens of publically available genome-wide datasets.	
GSE309660	The 7SK small nuclear ribonucleoprotein links the cell responses to transcription and replication stress by promoting replication fork reversal and homologous recombination	Many genotoxic agents that block replication fork progression to cause DNA replication stress also block transcription, which results in transcription stress. The 7SK-small nuclear ribonucleoprotein (7SK-snRNP) regulates and promotes RNA polymerase II activity in response to transcription	To inve campt hydrox we use to qua transc perfor osteos after 2 treatm before treatm

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		blocks. Individual 7SK-snRNP components HEXIM1, LARP7 and MEPCE are also reported to either facilitate or inhibit processes relevant to replication stress: conflicts between transcription and replication and homologous recombination (HR) at double-strand breaks (DSBs). Here, we investigate potential roles of 7SK-snRNP components in the response to replication stress-inducing agents such as camptothecin and hydroxyurea. We report that HEXIM1 and LARP7 promote replication fork slowing in response to agents that cause both replication- and transcription stress, in a manner consistent with their canonical 7SK-snRNP functions. Our data suggest that this role in fork slowing is mainly through facilitating RAD51-mediated replication fork reversal rather than transcription-replication conflicts. HEXIM1 and LARP7 promote RAD51 recruitment to replication-	contro gene e analysi from C

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		associated DSBs or post-replicative gaps under conditions of transcription stress such as induced by camptothecin or BET inhibitors, while to a lesser extent supporting HR at direct DSBs and not at all in absence of transcription stress. Our data support that LARP7 roles during replication stress are independent of its reported interaction with BRCA1 and both LARP7 and HEXIM1 promote survival of replication stress-inducing agents, especially camptothecin. 7SK-snRNP components are not recruited to stressed replication forks and RNA polymerase II inhibition phenocopies loss of these proteins. Taken together, our data support a model where 7SK-snRNP modulation of RNA polymerase II activity helps facilitate RAD51 function under transcription stress conditions, thereby linking the cell responses to transcription- and replication stress.	
GSE217544	The CDK12-INTAC Axis Modulates LEO1 Phosphorylation for Processive	Cyclin-dependent kinase 12 (CDK12) interacts with Cyclin K to form a a functional nuclear kinase complex, which has been	Detern Pol II o Detern synthe

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	Transcription Elongation	reported to phosphorylate the carboxyl-terminal domain (CTD) of RNA polymerase II (Pol II) for transcriptional regulation and co-transcriptional RNA processing. However, the precise mechanisms and targets of CDK12 action remain largely unknown. Here, we combined a chemical genetic screen and phosphoproteomic strategies and identified a landscape of nuclear CDK12 substrates, which included proteins that regulate transcription, chromatin organization, and RNA splicing. Next, we confirmed that the LEO1 subunit of the transcription elongation factor PAF1 complex (PAF1C) is a bona fide substrate of CDK12. Acute depletion of LEO1 reduces Pol II occupancy on the chromatin, while mutations of LEO1 phosphorylation sites to non-phosphorylatable alanine residues attenuated the association of PAF1C with elongating Pol II and chromatin, resulting in impaired processive transcription elongation. Furthermore,	Determ Pol II b

GSE ID	Title	Summary	Overa
		LEO1 C-terminus could interact with and be dephosphorylated by the Integrator-PP2A complex (INTAC), while acute depletion of INTAC in cells promotes the association between PAF1C and elongating Pol II on the chromatin. Together, this study provides a novel transcriptional regulatory mechanism that the CDK12-INTAC axis fine-tunes LEO1 phosphorylation for processive transcription elongation.	
GSE139463	Selective Mediator-dependence of cell type-specifying transcription [TT-seq]	The Mediator complex directs signals from DNA-binding transcription factors to RNA polymerase (Pol) II. Despite this pivotal position, mechanistic understanding of Mediator in human cells remains incomplete. Here, we quantified Mediator-controlled Pol II kinetics by coupling rapid subunit degradation with orthogonal experimental readouts. Consistent with a model of condensate-driven transcription initiation, large clusters of hypo-phosphorylated Pol II rapidly disassembled upon Mediator degradation. This	TT-seq dTAG dTAG7 NVP2 MED1 500nM

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		was accompanied by a selective and pronounced disruption of cell type-specifying transcriptional circuits, whose constituent genes featured exceptionally high rates of Pol II turnover. Notably, transcriptional output of most other genes was largely unaffected by acute Mediator ablation. Maintenance of transcriptional activity at these genes was linked to an unexpected, CDK9-dependent compensatory feedback loop that elevated Pol II pause release rates genome-wide. Collectively, our work positions human Mediator as a globally acting coactivator that selectively safeguards the functionality of cell type-specifying transcriptional networks.	
GSE161684	Role of hNABP2 (hSSB1) in transcriptional regulation	Use Global Run-On assay (GRO)-seq, chromatin immunoprecipitation (ChIP)-seq, and DNA: RNA immunoprecipitation (DRIP)-seq to investigate whether and how hNABP2 (hSSB1) regualte transcription.	Global Seq: na down b beads, the GR conver library NEBNe library ChIP-S crossli DNA w

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			sonica DNA w hNABF DNA w NEBNe buffer adaptc 0.9 AM adaptc amplifi 24, 26 Multipl and ur NEB) f experi immun Seq: TI immun perfor with S was ex sample extrac The lib using prep k
GSE113171	Mapping multi-layered regulation in response to environmental stress	We have mapped transcript and protein concentrations in cervical cancer cells responding to treatment with tunicamycin over an eight hour time course. We complement this dataset with RNA footprints of ribosomes and non-ribosomal proteins to determine changes in translation and trans-factor binding. The goals	Hela c ER str Transc and pr were g time p hours) triplica Illumin

GSE ID	Title	Summary	Overa
		of this study are to unveil the time dependent complexity of the signaling pathways at the level of transcription and translation during different stress conditions like stress of the endoplasmic reticulum and oxidative stress.	
GSE267044	Disruption of the Integrator complex triggers the Integrated Stress Response through formation of double-stranded RNA [RIPseq]	Acute loss of INT induces transcription factors that initiate stress-responses like the Integrated Stress Response (ISR). Inefficient elongation upon INT loss enables termination in introns and dsRNA formation. dsRNA is recognized by PKR, which persistently activates the ISR in samples from patients with INT mutations	RIP-se antibo absen of INT
GSE294088	NDF/GLYR1 Promotes RNA Polymerase II Processivity via Pol II Binding and Nucleosome Destabilization	The Nucleosome Destabilizing Factor (NDF) facilitates transcription through chromatin, but its precise mechanism remains incompletely understood. Here, we identify a critical region (amino acids 140-160) within NDF that specifically interacts with phosphorylated RPB1, the largest subu-nit of elongating RNA Polymerase II (Pol II).	TT-seq RNA a NDF W cells w and re Bru-se RNA ir rescue MT4.

GSE ID	Title	Summary	Overa
		Mutations in this region disrupt Pol II interaction and impair Pol II elongation both in vitro and in cells, yet do not affect NDF's ability to destabilize nu-cleosomes, establishing a functional separation between these two activities. Cellular studies re-veal that NDF knockout cells display faster Pol II elongation rates but produce fewer nascent tran-scripts, demonstrating NDF's primary role in maintaining transcriptional processivity through-out gene bodies. Our findings demonstrate that NDF uses distinct mechanisms to ensure productive transcription elongation rather than simply enhancing elongation speed, offering new insights in-to how transcription efficiency is maintained in chromatin.	
GSE262518	Acute multi-level response to defective de novo chromatin assembly in S-phase	Long-term perturbation of de novo chromatin assembly during DNA replication has profound effects on epigenome maintenance and cell fate. The early mechanistic origin of these defects is unknown. Here, we combine acute degradation of Chromatin	RPE1 c treatec (ctrl) o CHAF1 disrup assem then s seq, V experi chrom

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		Assembly Factor 1 (CAF-1), a key player in de novo chromatin assembly, with single-cell genomics, quantitative proteomics, and live-microscopy to uncover these initiating mechanisms in human cells. CAF-1 loss immediately slows down DNA replication speed and renders nascent DNA hyper-accessible. A rapid cellular response, distinct from canonical DNA damage signaling, is triggered and lowers histone mRNAs. As a result, histone variants usage and their modifications are altered, limiting transcriptional fidelity and delaying chromatin maturation within a single S-phase. This multi-level response induces a cell-cycle arrest after mitosis. Our work reveals the immediate consequences of defective de novo chromatin assembly during DNA replication, explaining how at later times the epigenome and cell fate can be altered.	transcr replica

GSE ID	Title	Summary	Overa
GSE138356	Enhancers predominantly control transcription initiation [mNET-seq]	Gene transcription occurs via a cycle of linked events including initiation, promoter proximal pausing and elongation of RNA polymerase II (Pol 2). A key question is how do transcriptional enhancers influence these events to control gene expression? Here we have used a new approach to quantify transcriptional initiation and pausing in-vivo, while simultaneously identifying transcription start sites (TSSs) and pause-sites (TPSs). When analysed in parallel with nascent RNA-seq, these data show that differential gene expression is achieved predominantly via changes in transcription initiation rather than Pol 2 pausing or elongation. Using genetically engineered mouse models deleted for specific enhancers we show that these elements control gene expression via Pol 2 recruitment and/or initiation rather than via promoter proximal pause release. Using genome-wide analysis we show that enhancers, in general, control gene expression at the stage of Pol 2	To cha engage perfor cells, p cells fr R1R2 I genoty enhanc homoz combi A com deletio (a.k.a - transc the alp report Nature

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		recruitment and initiation rather than via pausing.	

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